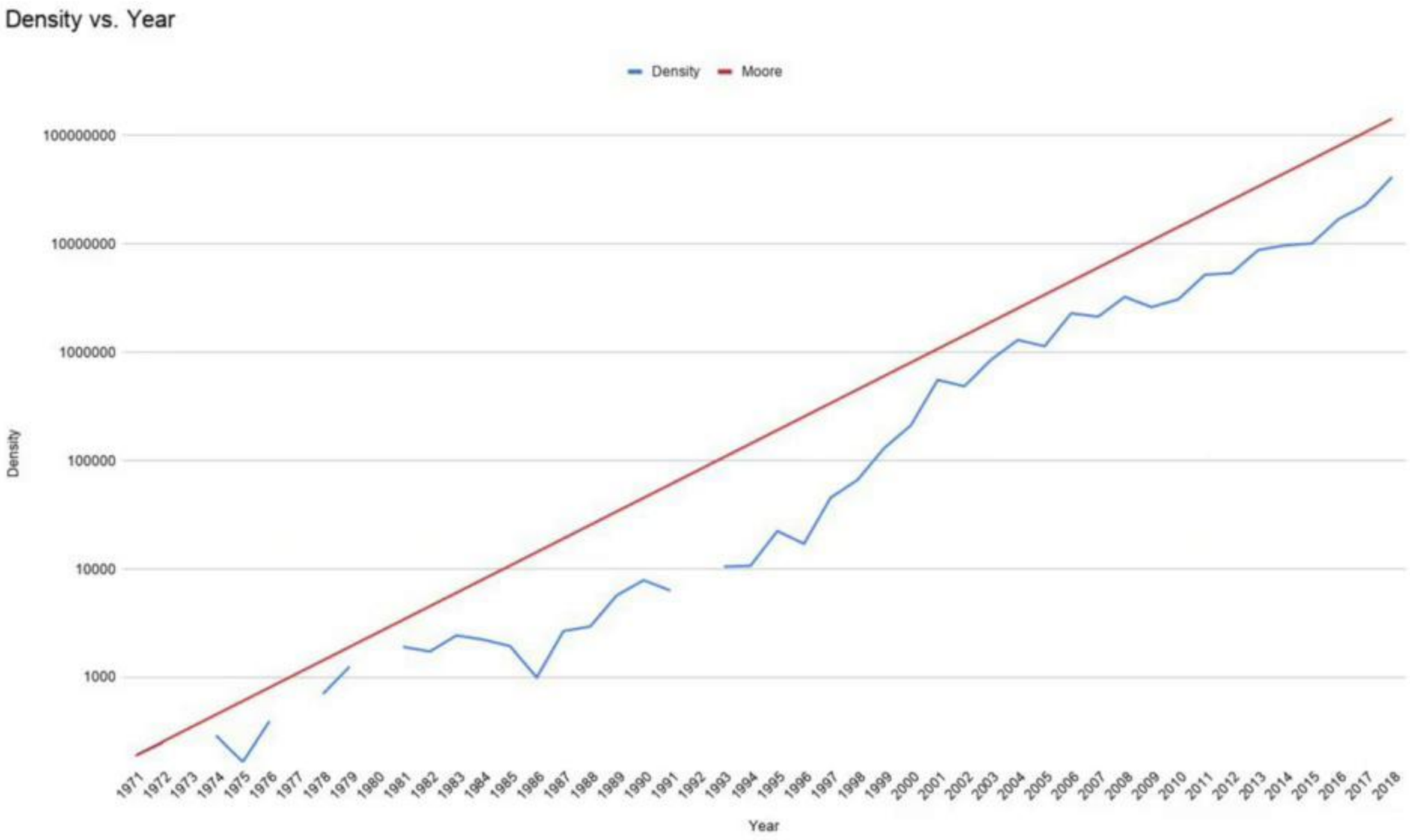
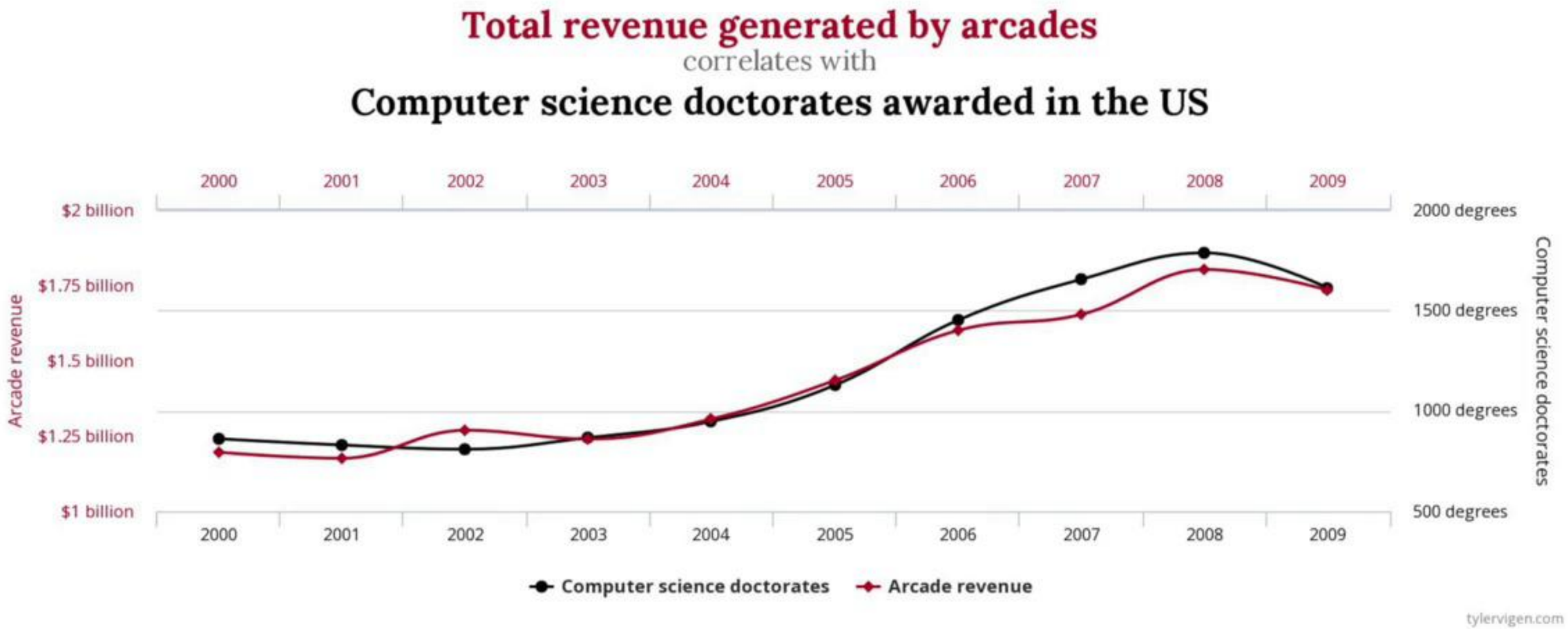
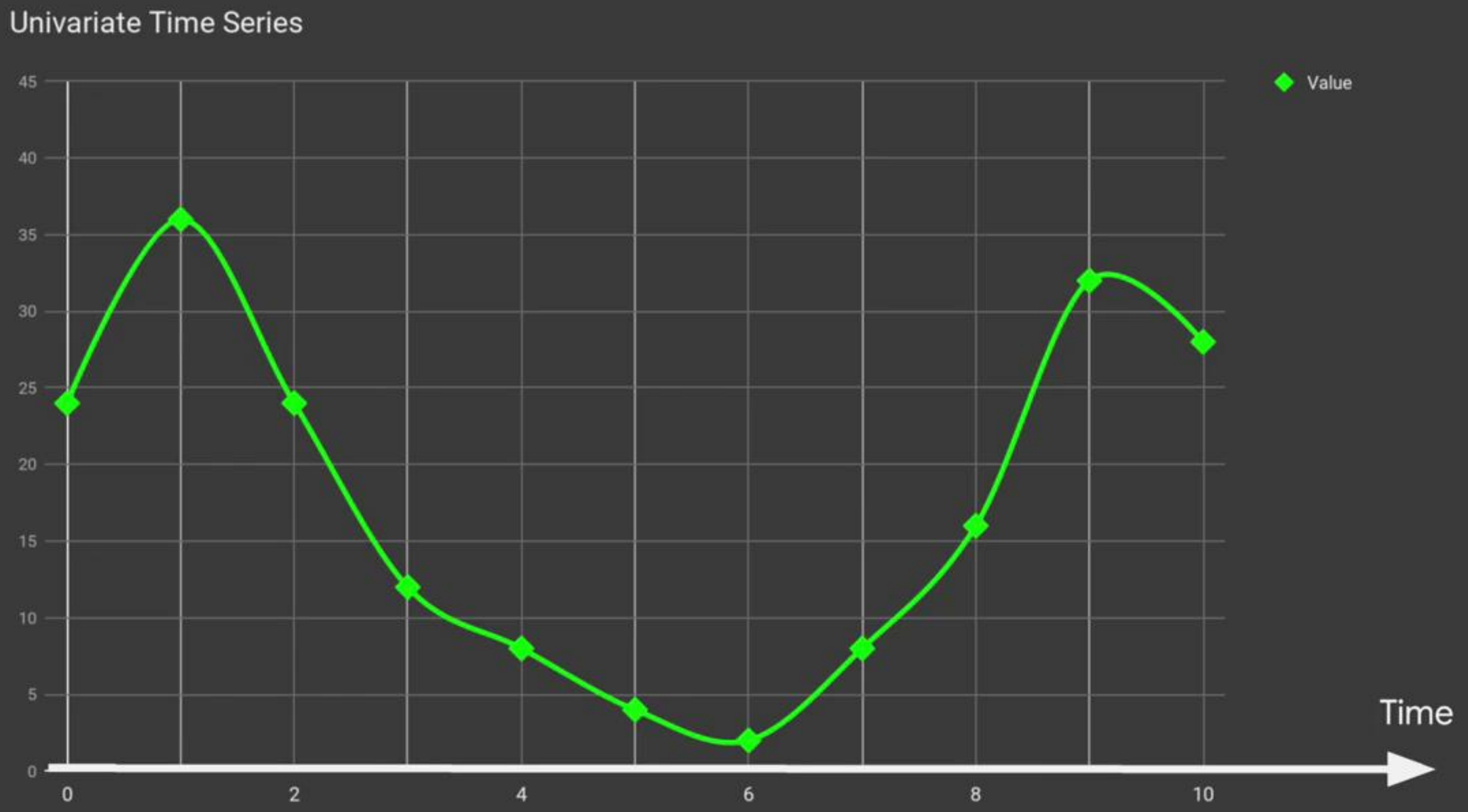


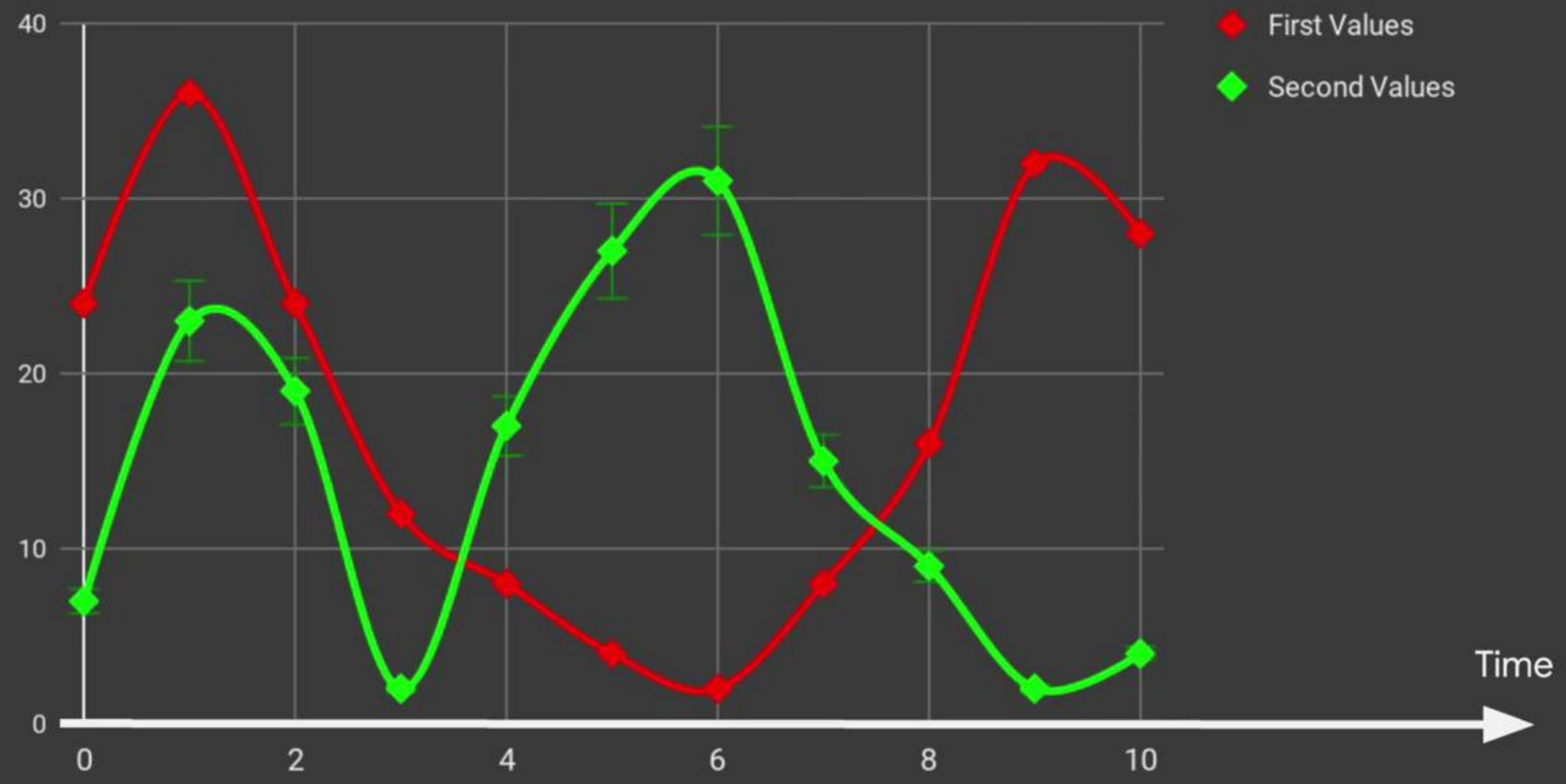
Chart Created by lmoroney@

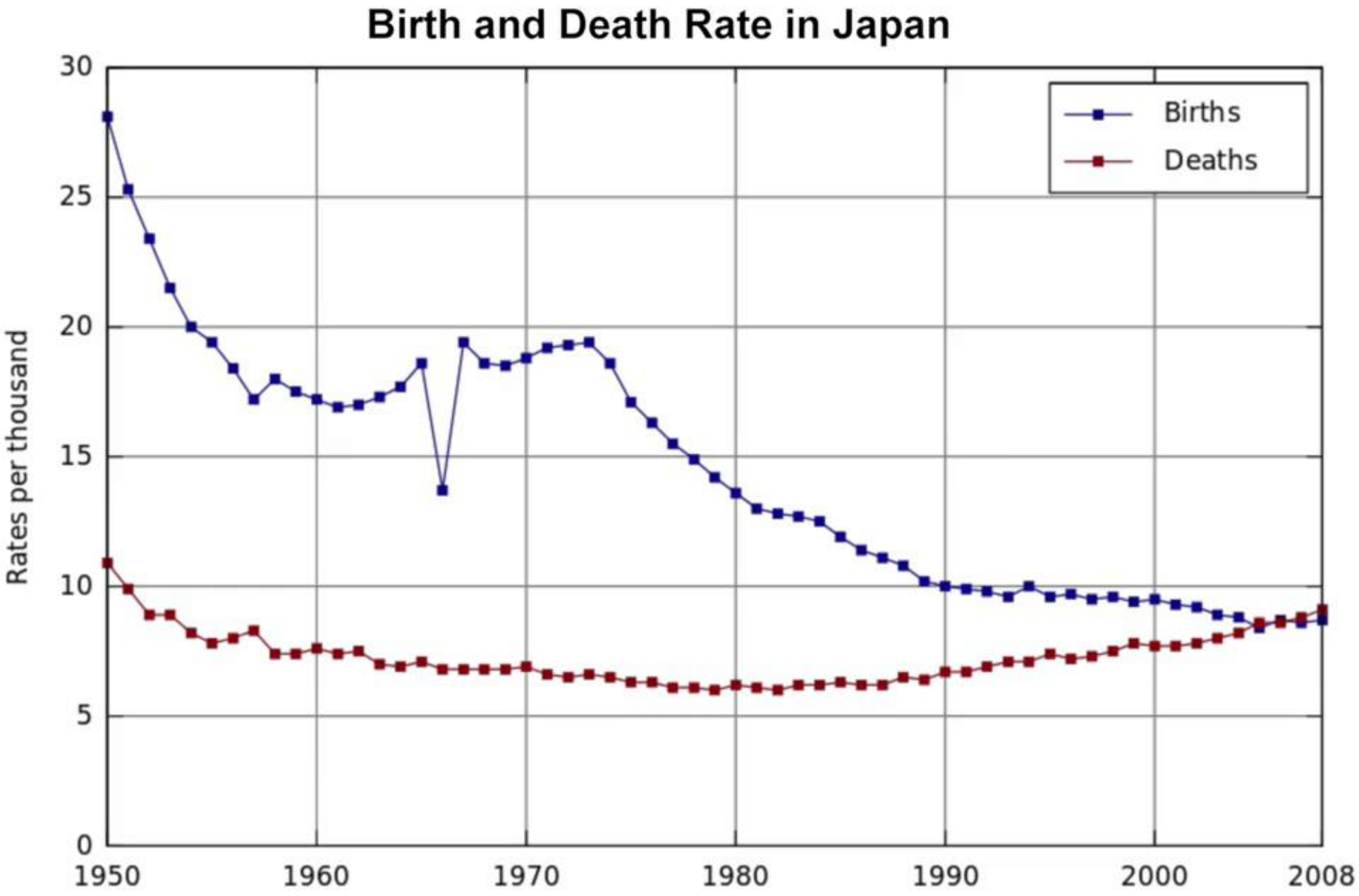


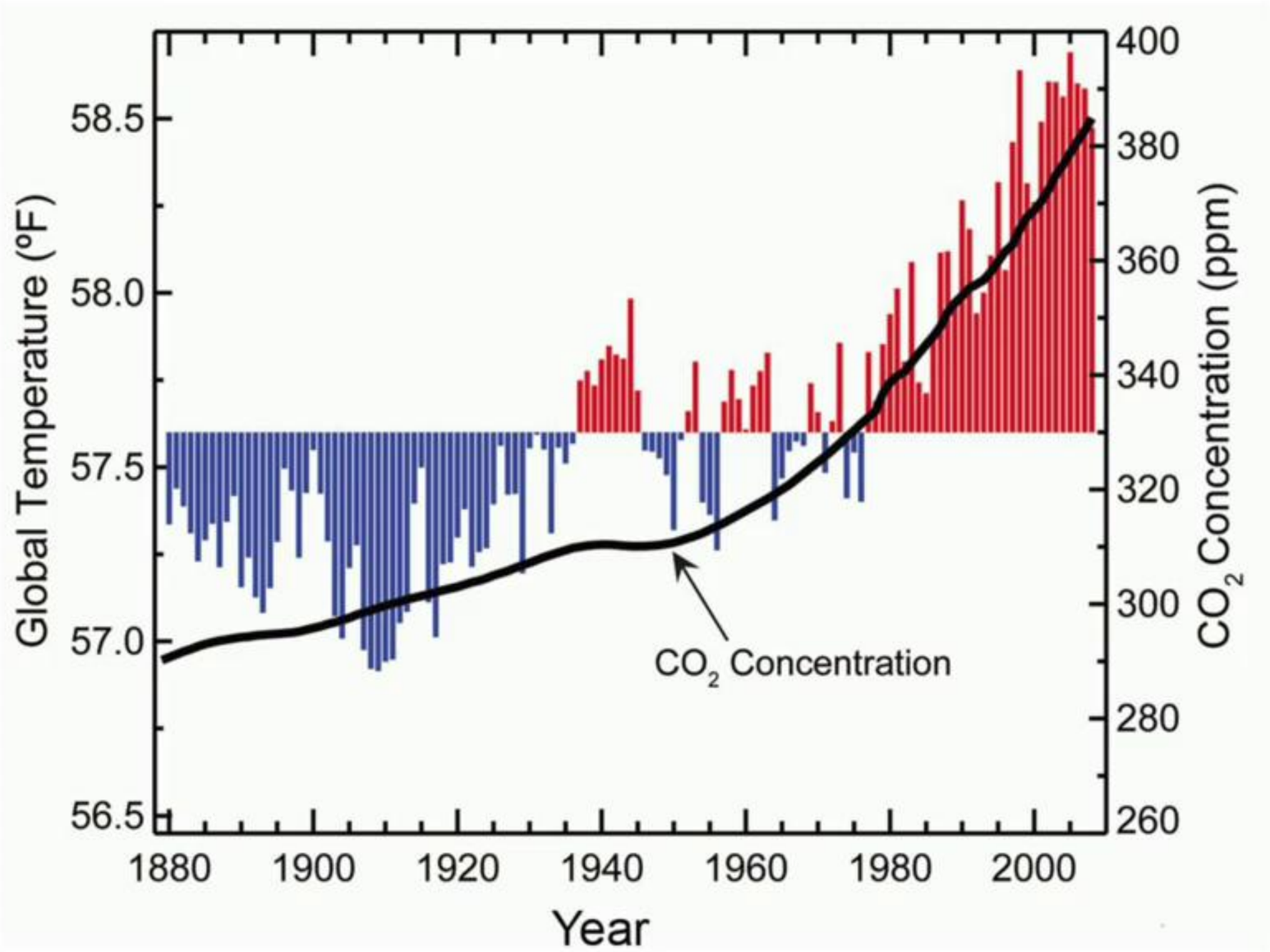
CC By 4.0 - TylerVigen.com

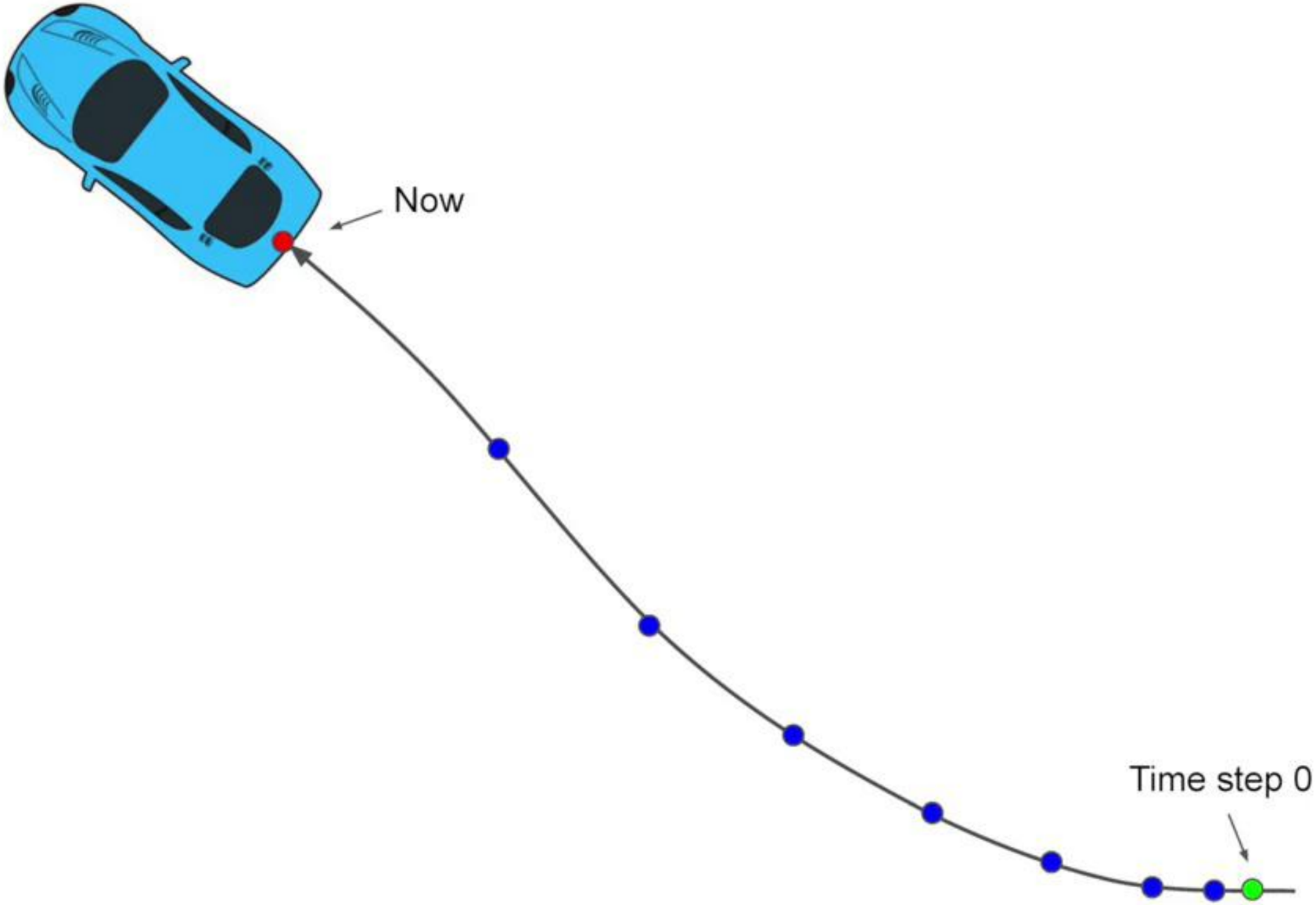

Univariate Time Series
Value
45
40
35
30
25
20
15
10
5
0
0
2
4
6
8
10
Time

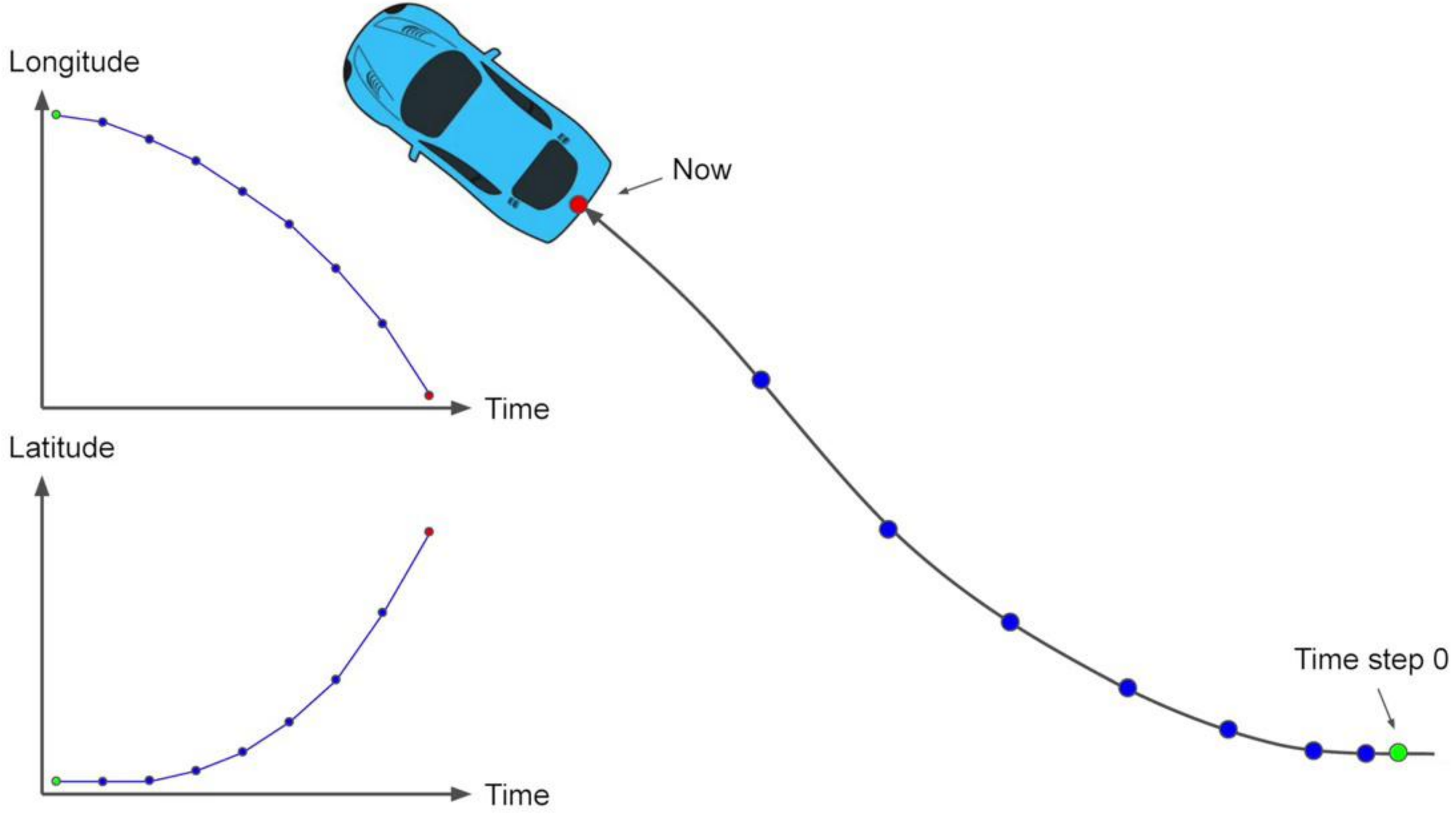
Multivariate Time Series



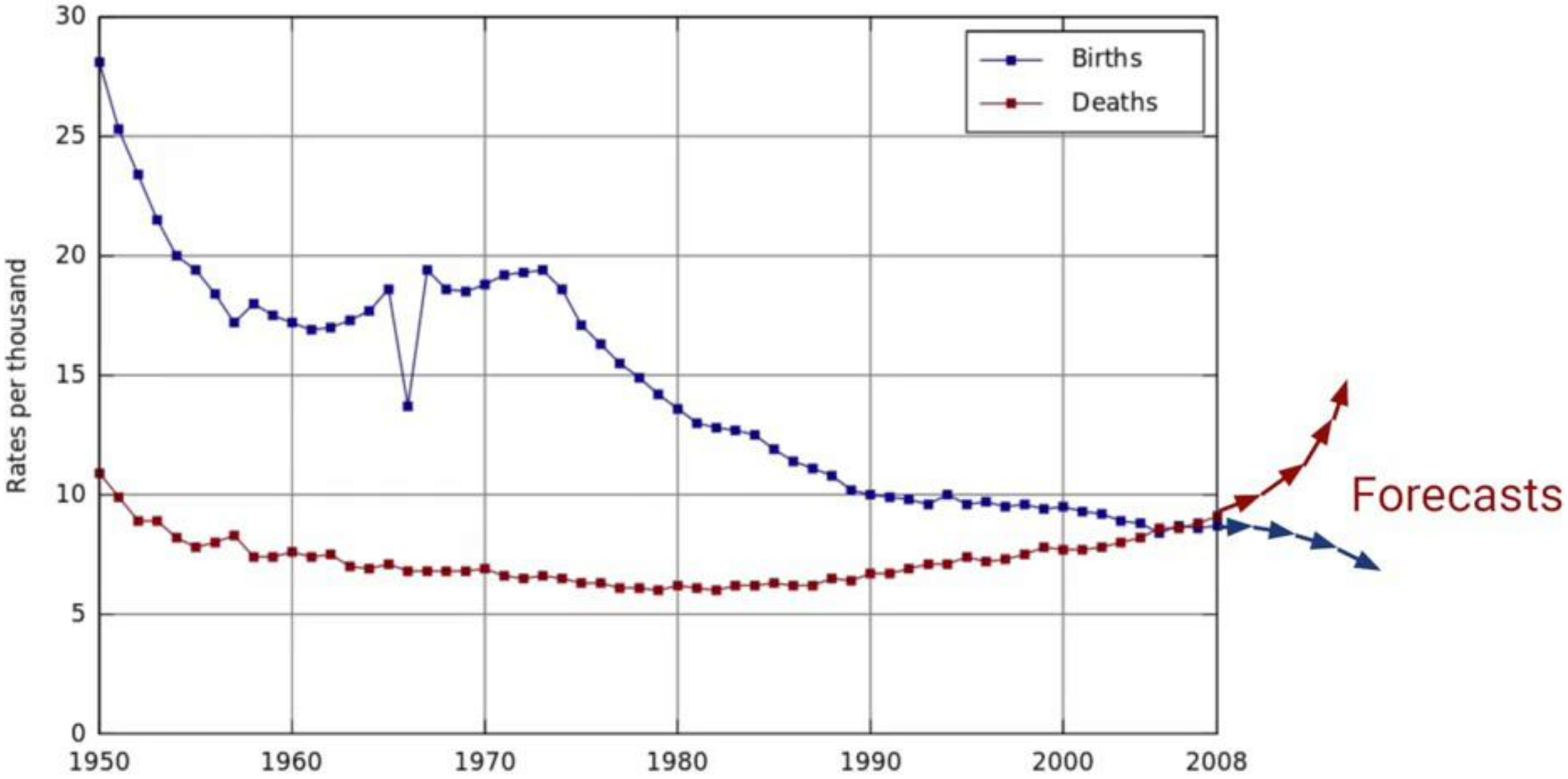

Birth and Death Rate in Japan
Births
Deaths
Rates per thousand
30
25
20
15
10
5
0
1950
1960
1970
1980
1990
2000
2008


Global Temperature (ºF)
58.5
58.0
57.5
57.0
56.5
400
380
360
340
320
300
280
260
CO_2 Concentration (ppm)
CO_2 Concentration
1880
1900
1920
1940
1960
1980
2000
Year


Now
Time step 0


Longitude
Time
Latitude
Time
Now
Time step 0

Birth and Death Rate in Japan



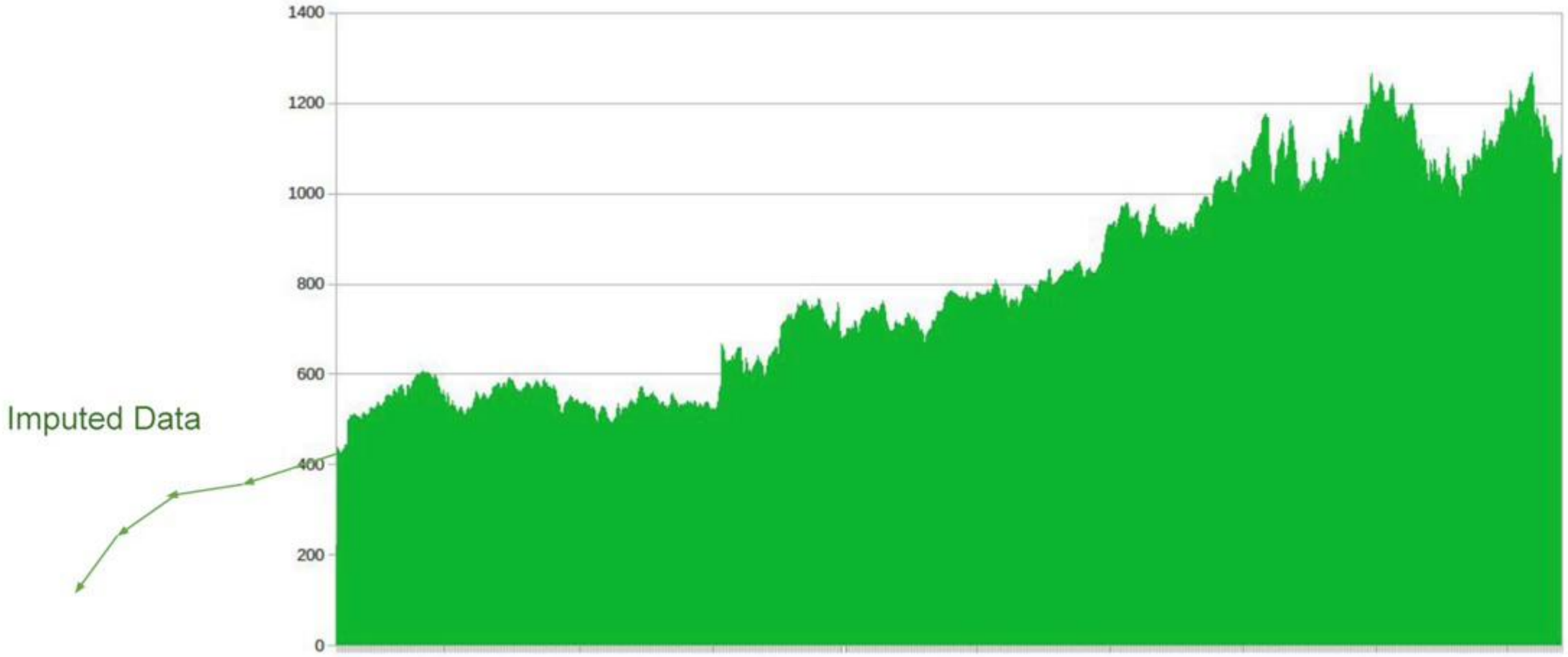


Chart Created by lmoroney@

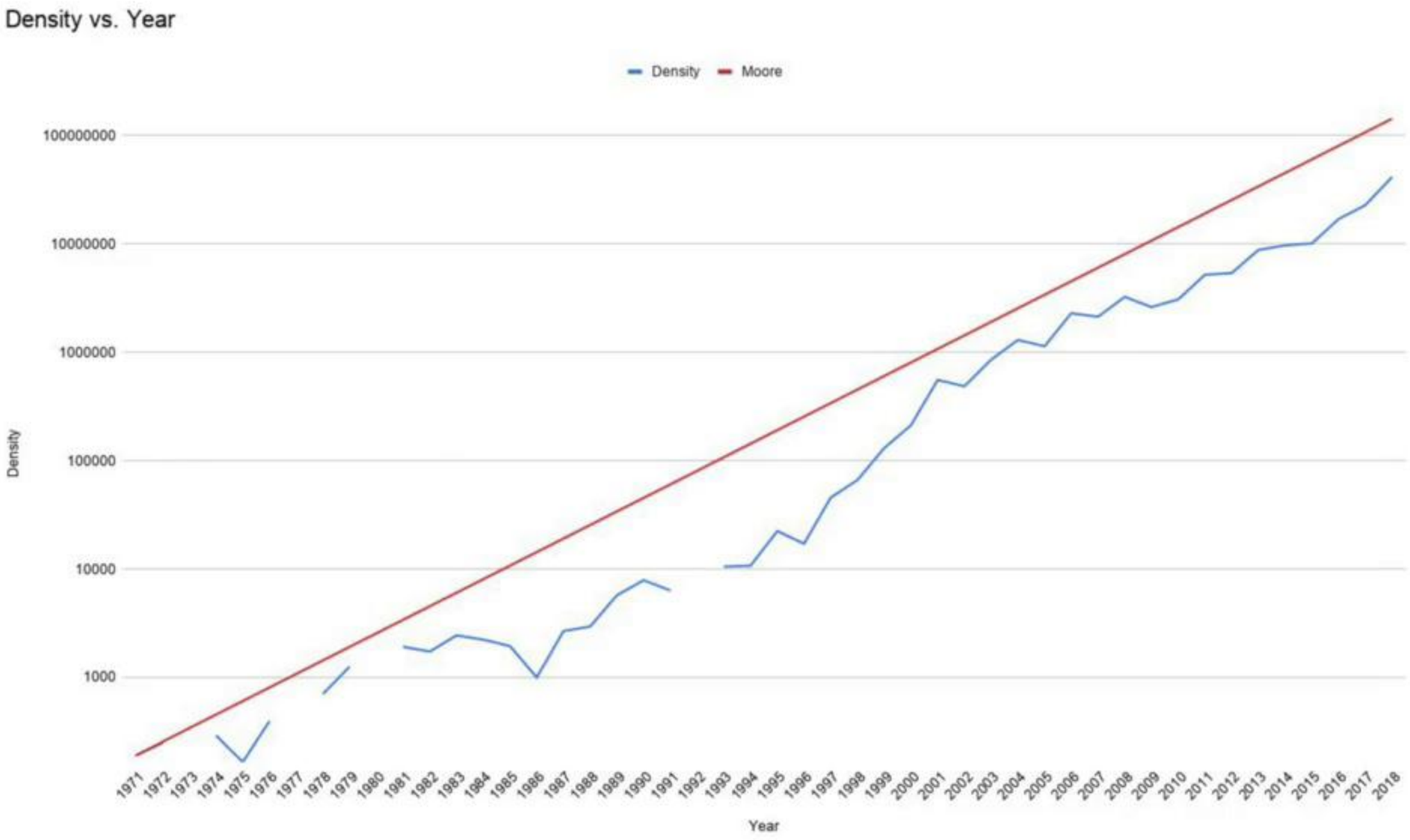
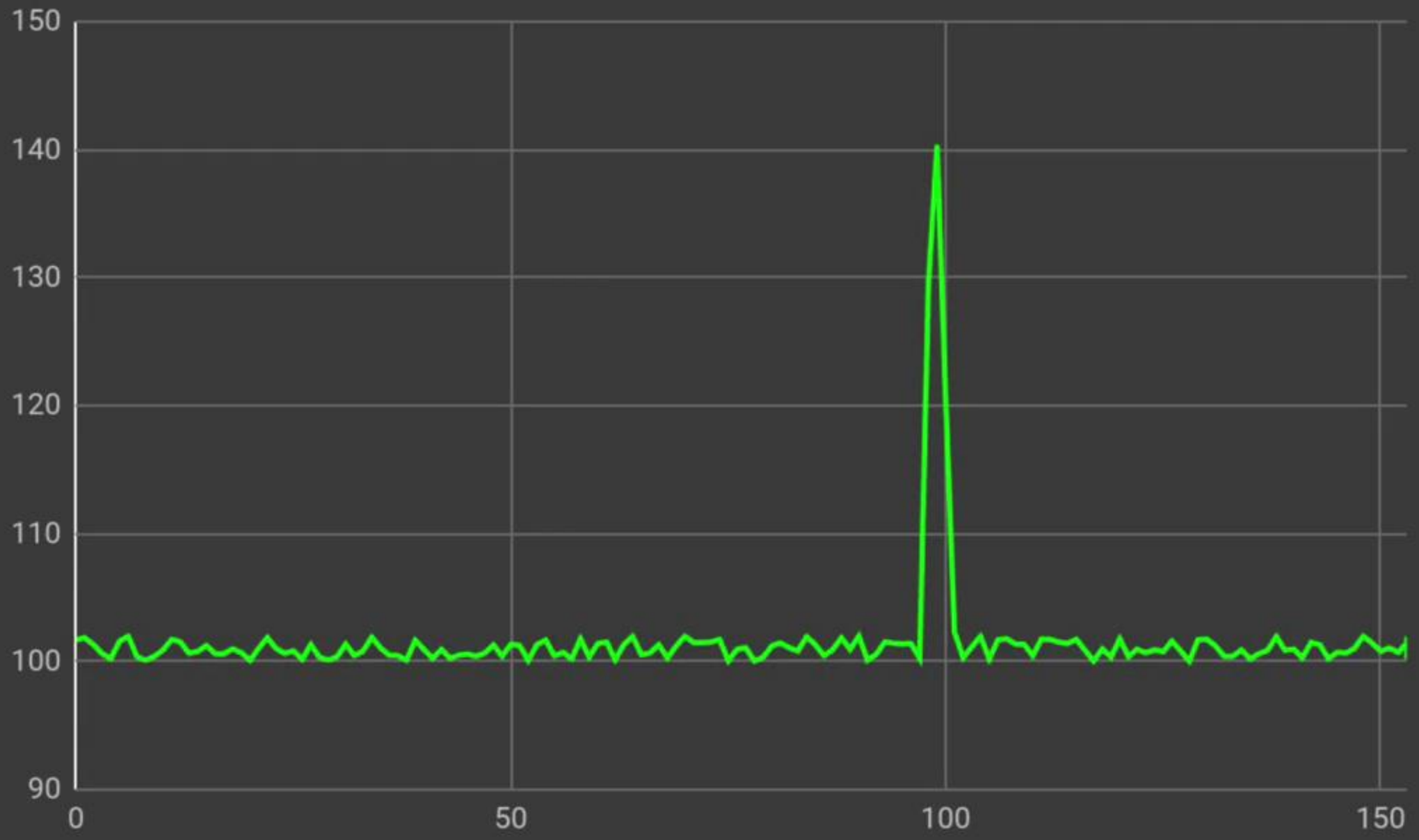
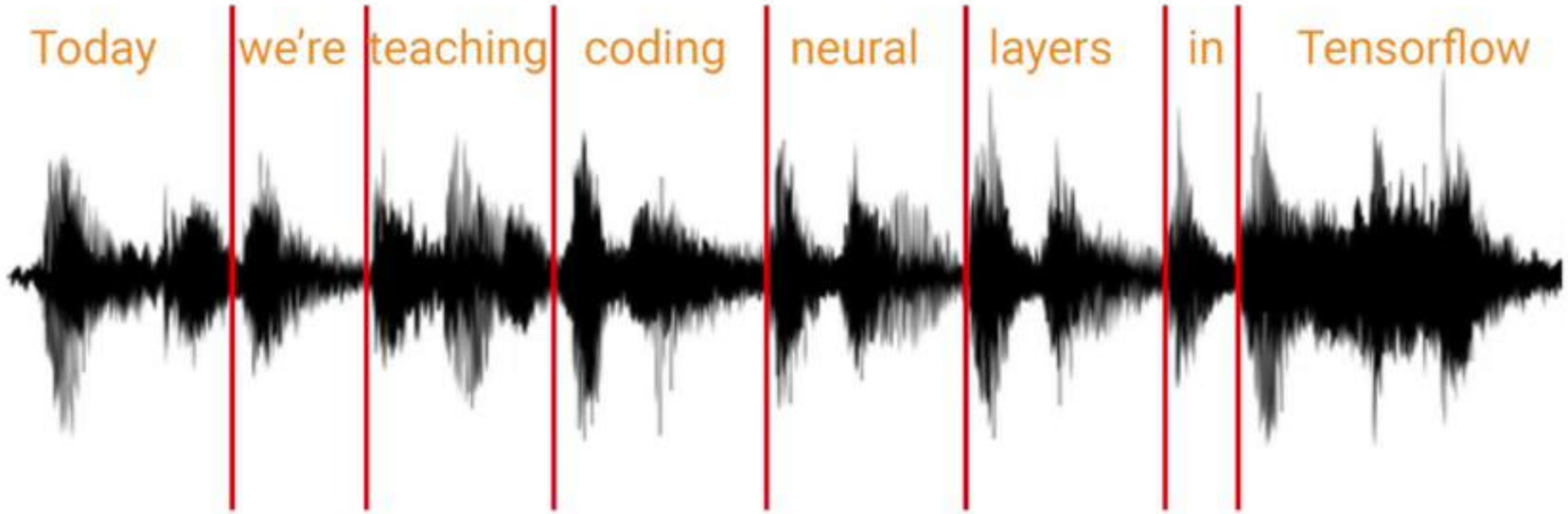


Chart Created by lmoroney@


150
140
130
120
110
100
90
0
50
100
150


Today
we're
teaching
coding
neural
layers
in
Tensorflow

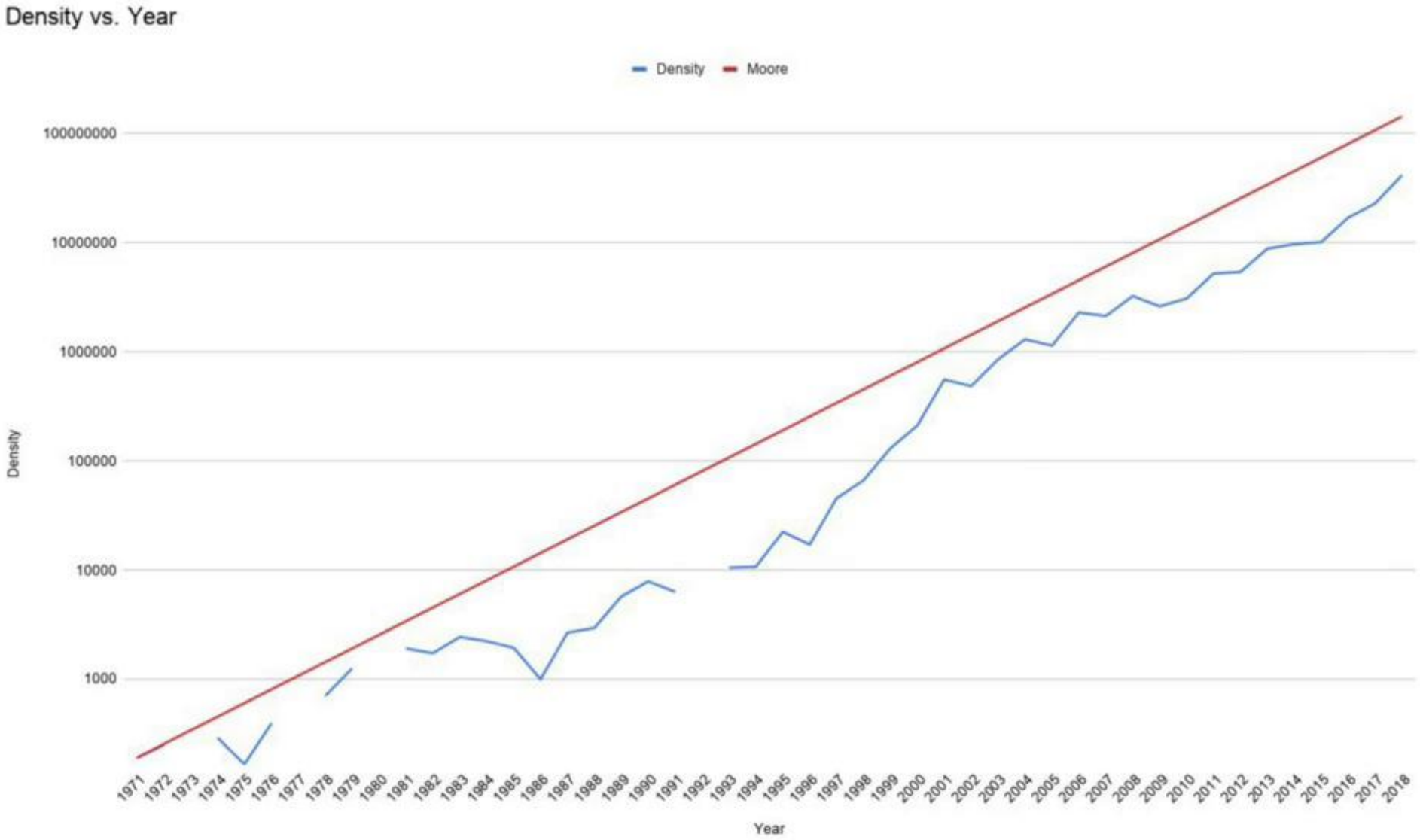
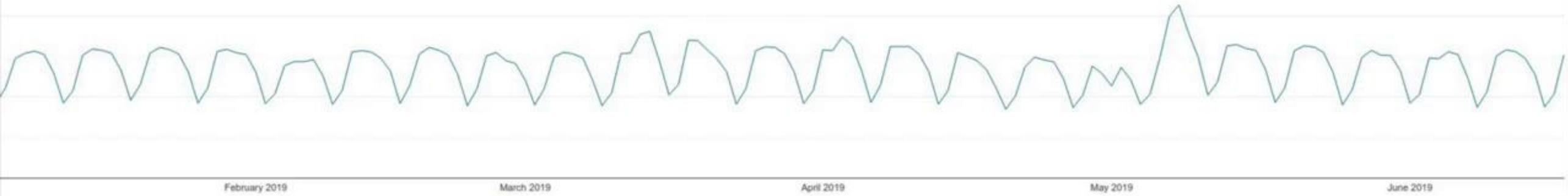
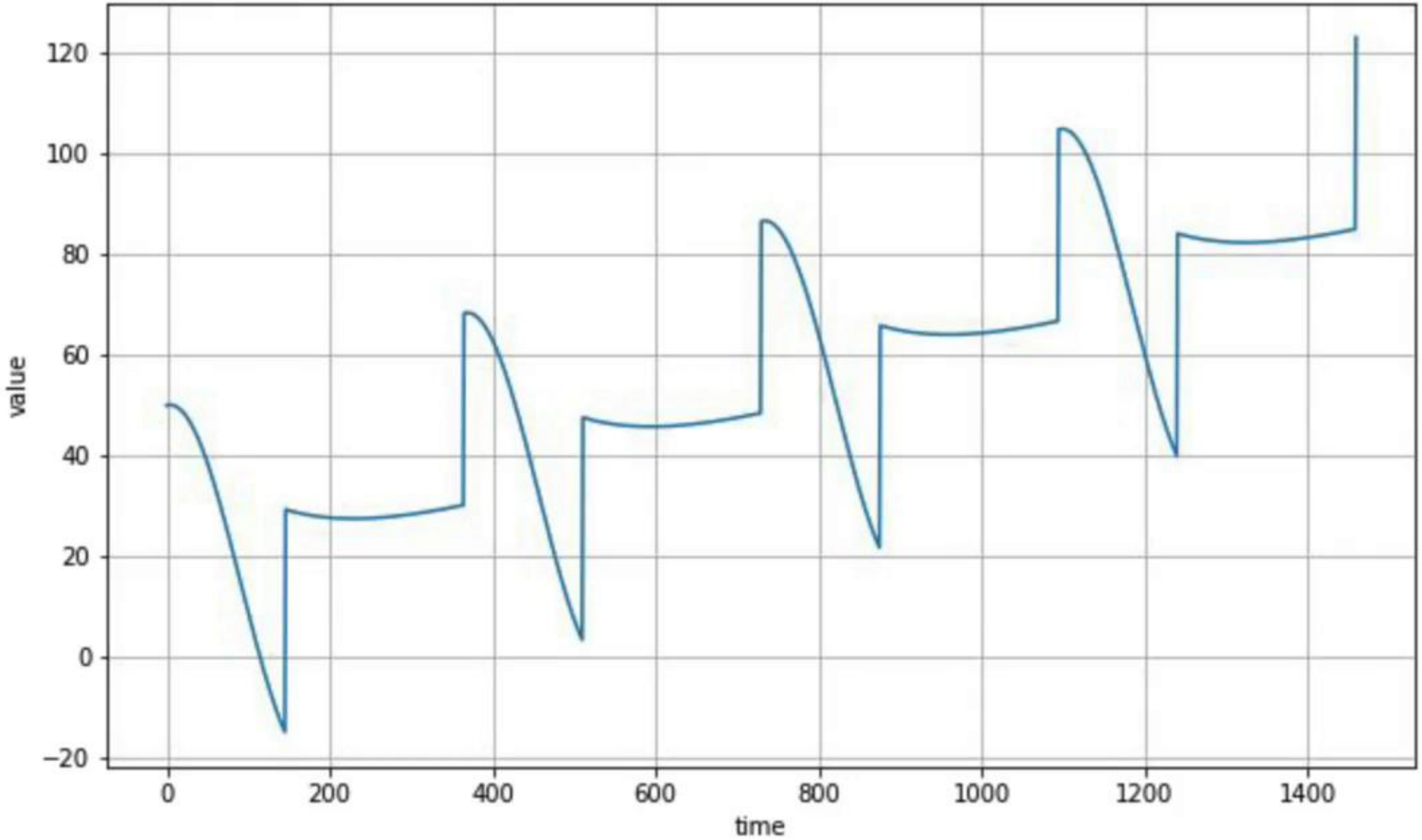
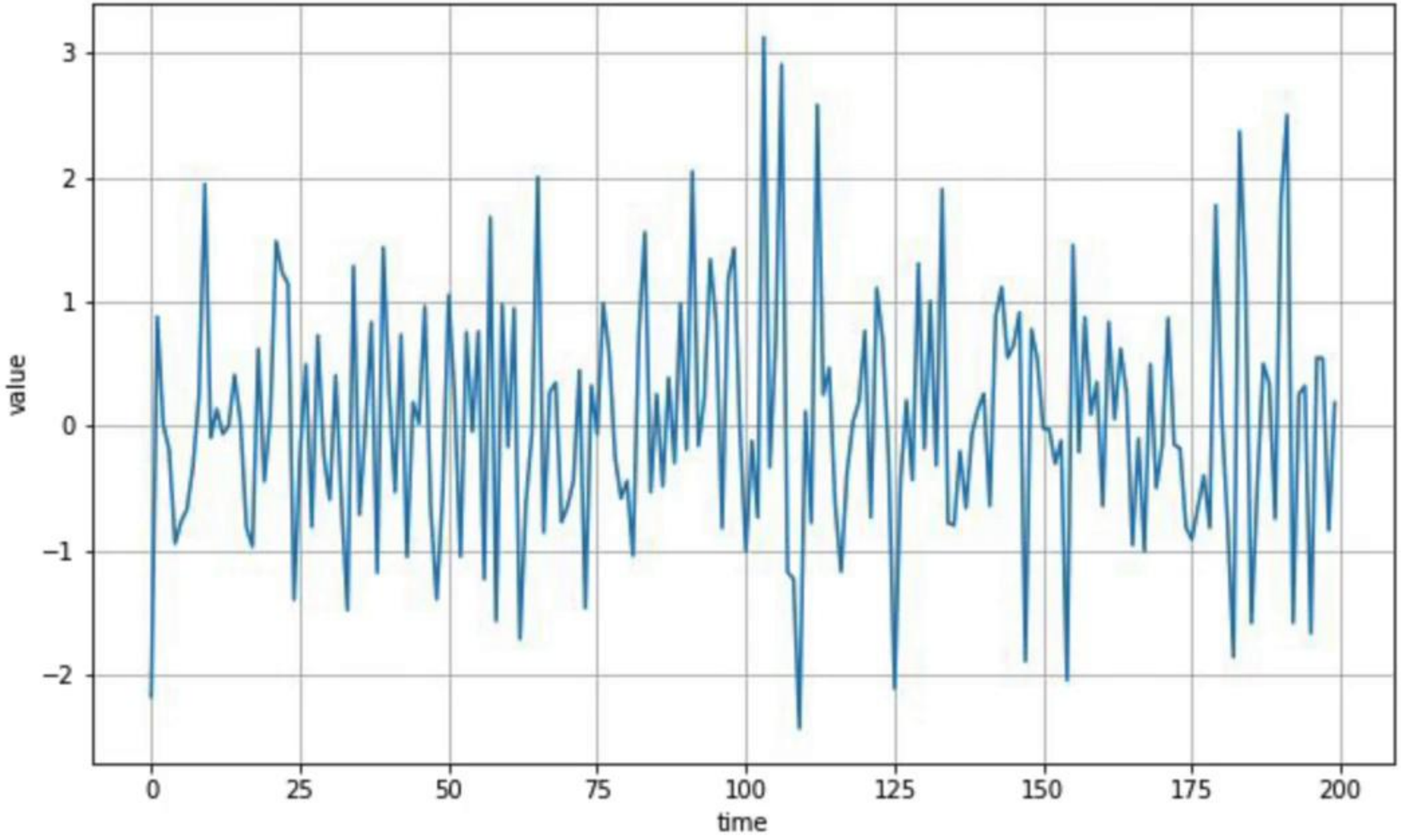


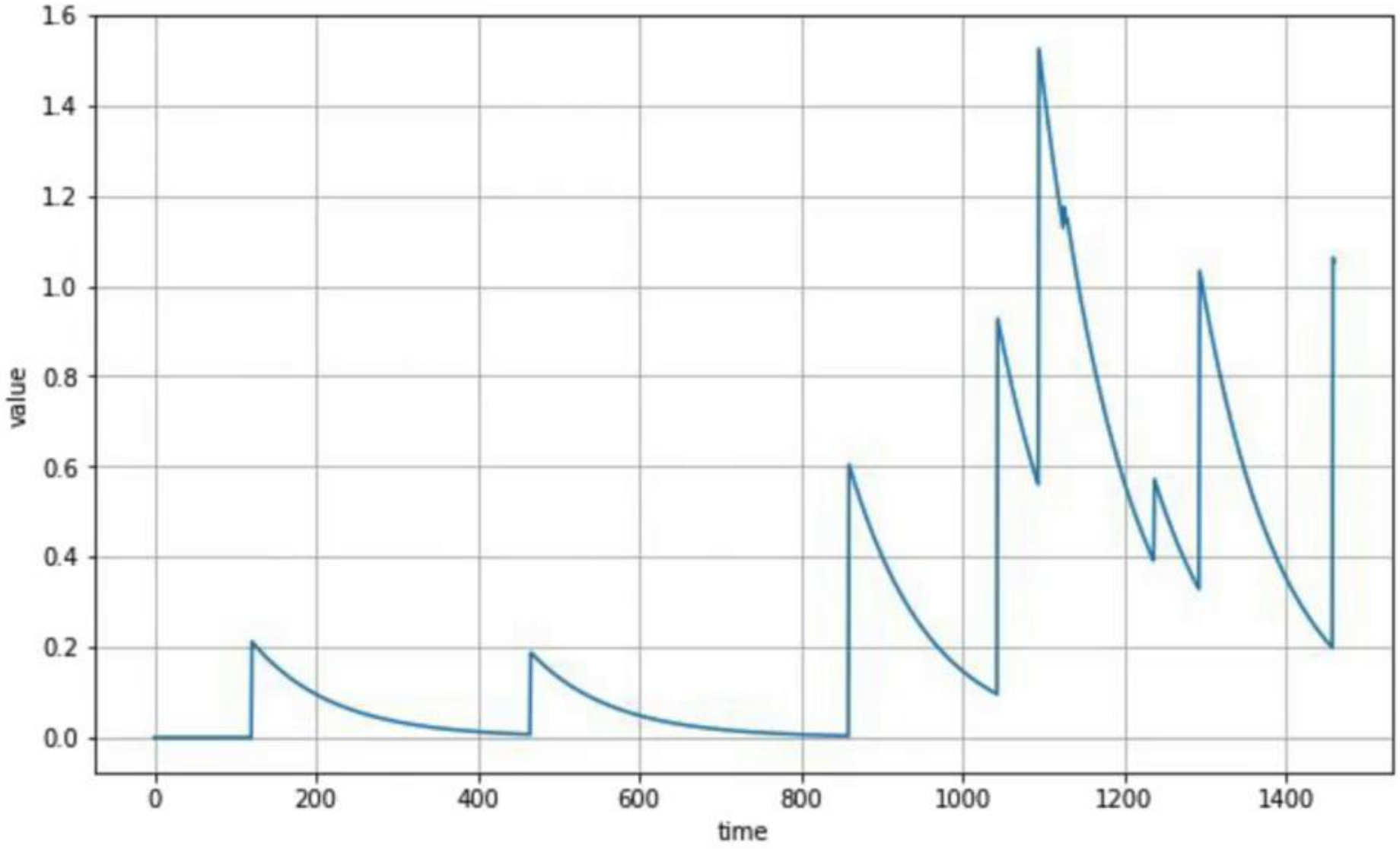
Chart Created by lmoroney@


February 2019
March 2019
April 2019
May 2019
June 2019

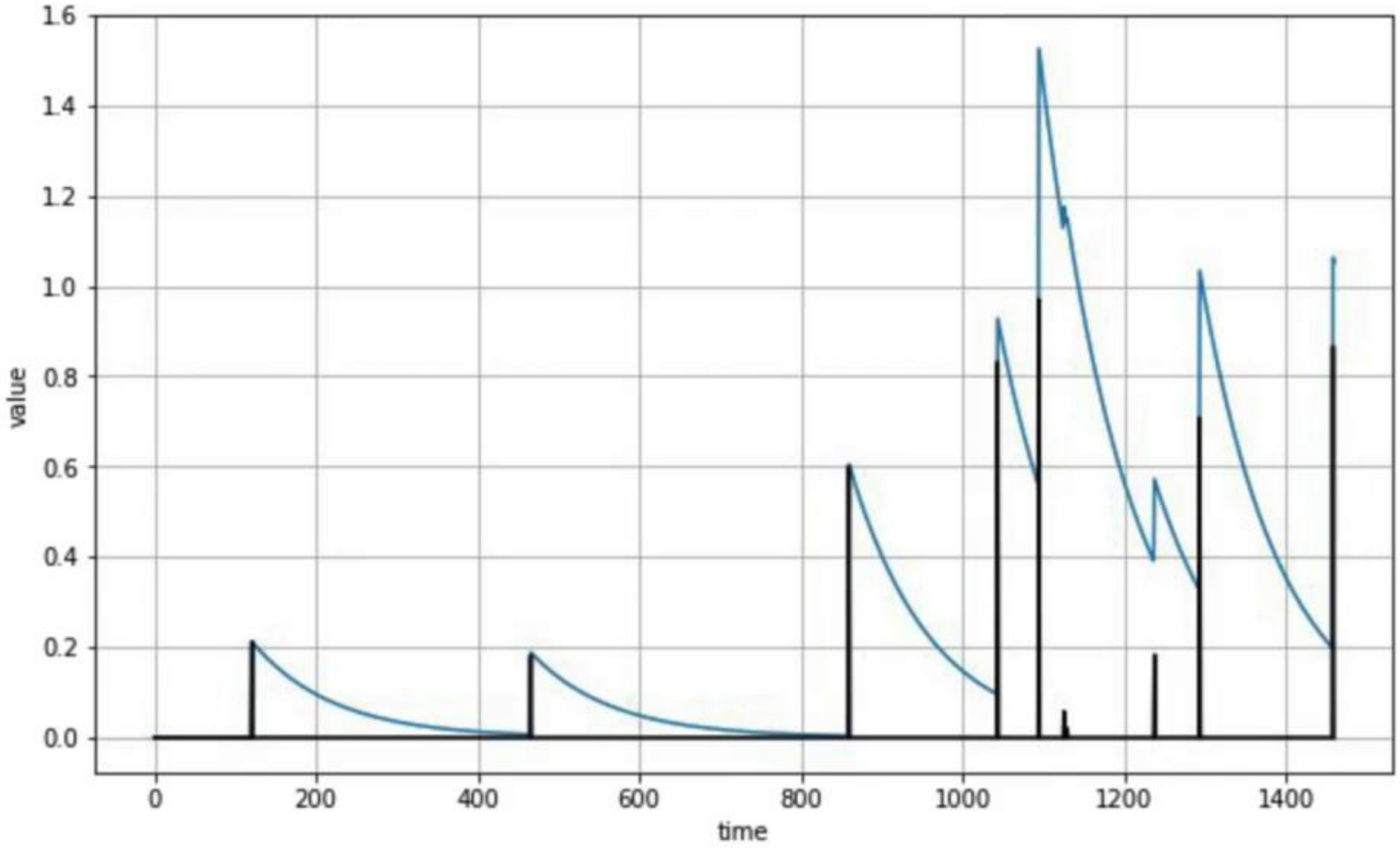

120
100
80
60
40
20
0
−20
value
0
200
400
600
800
1000
1200
1400
time


3
2
1
0
−1
−2
value
0
25
50
75
100
125
150
175
200
time

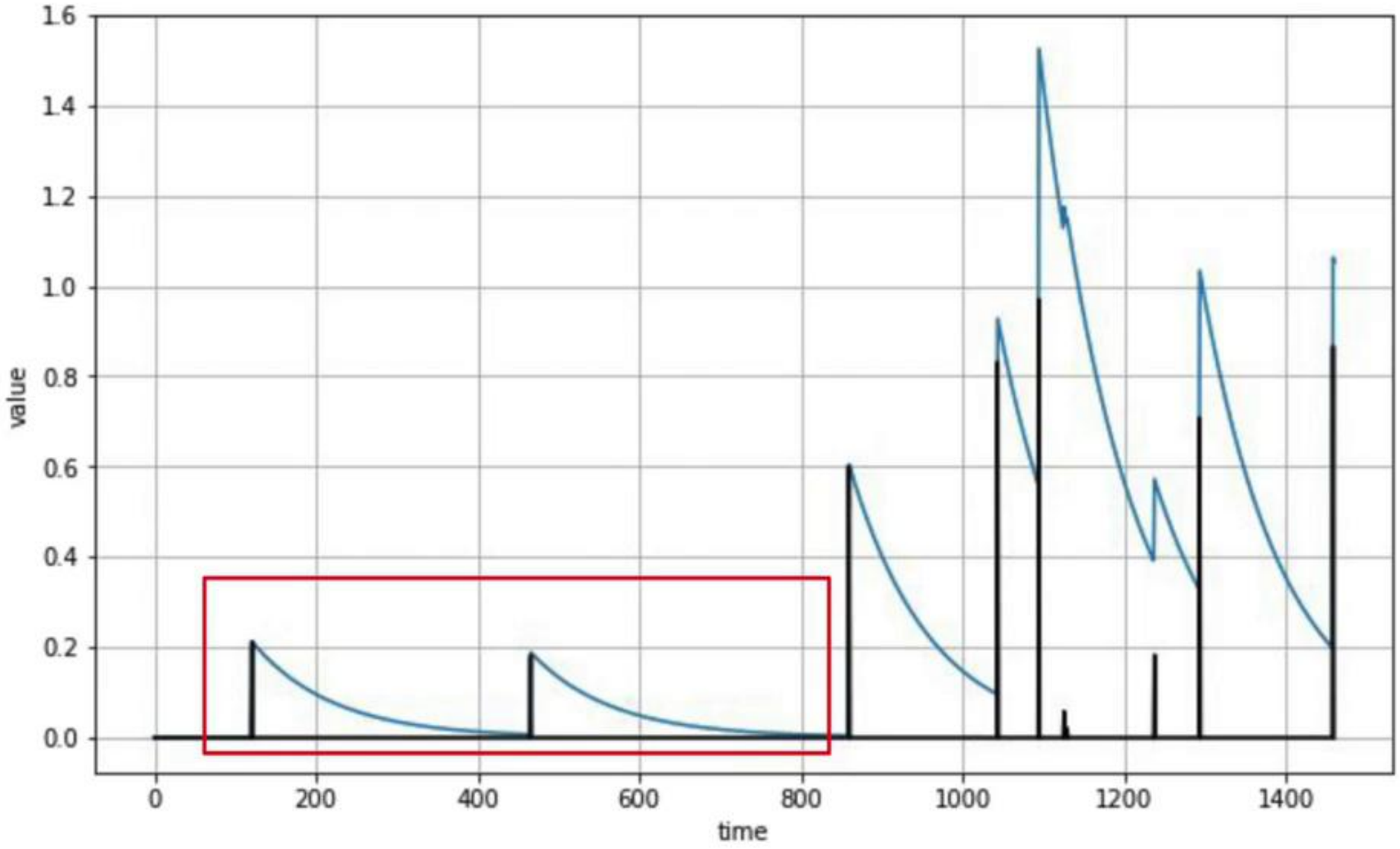
Autocorrelation



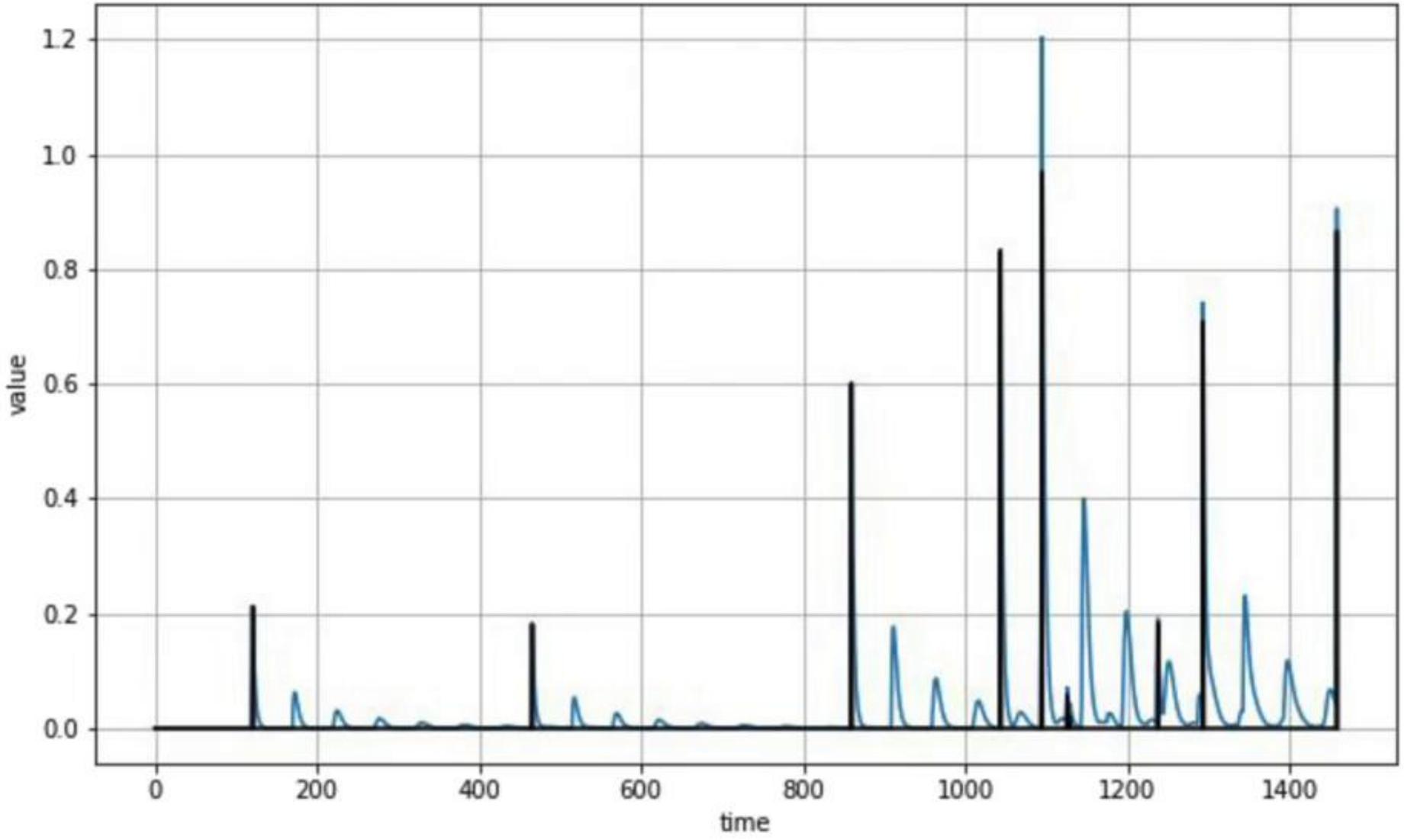
$v(t) = 0.99 \times v(t-1)$ + occasional spike



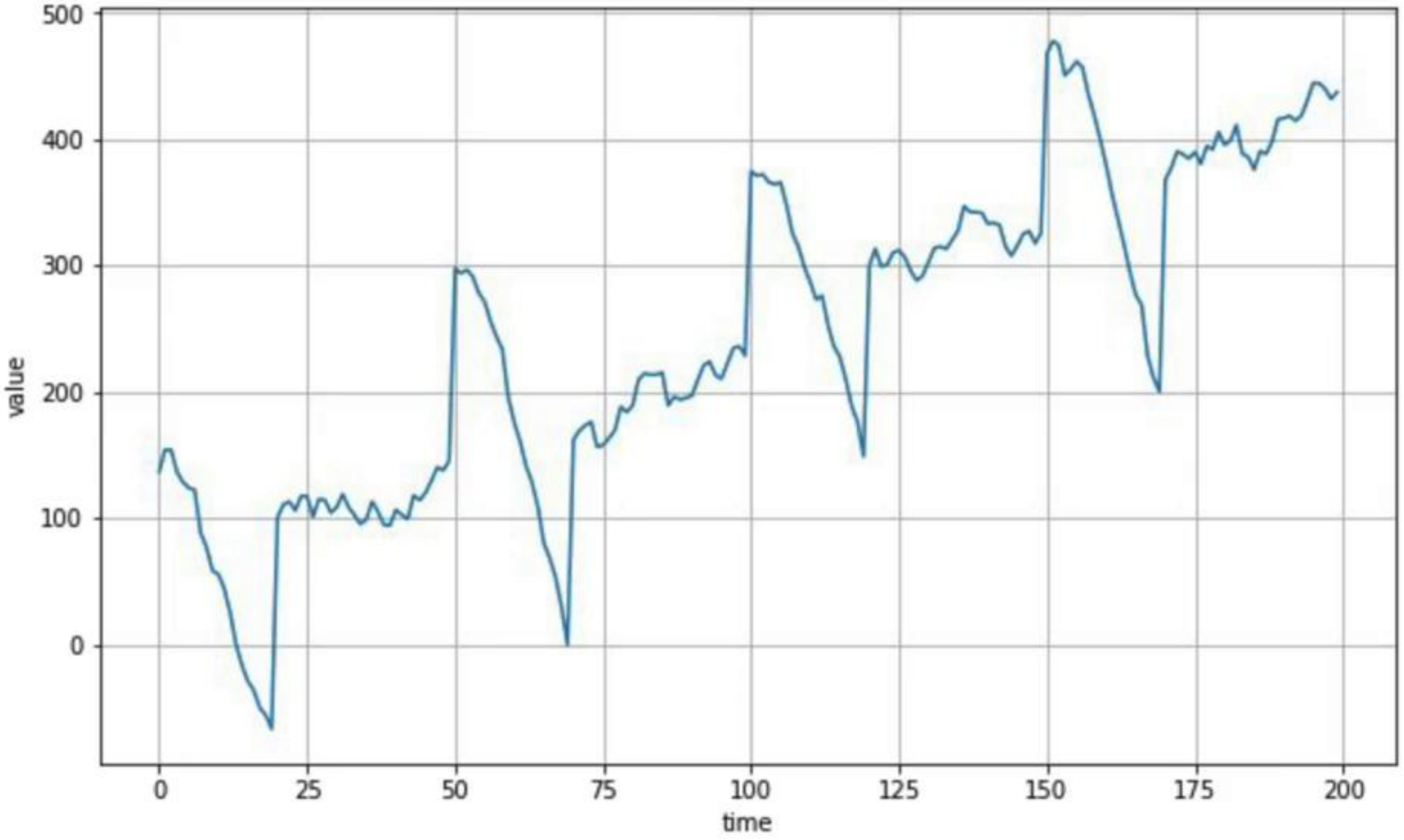
$v(t) = 0.99 \times v(t-1)$ + occasional spike



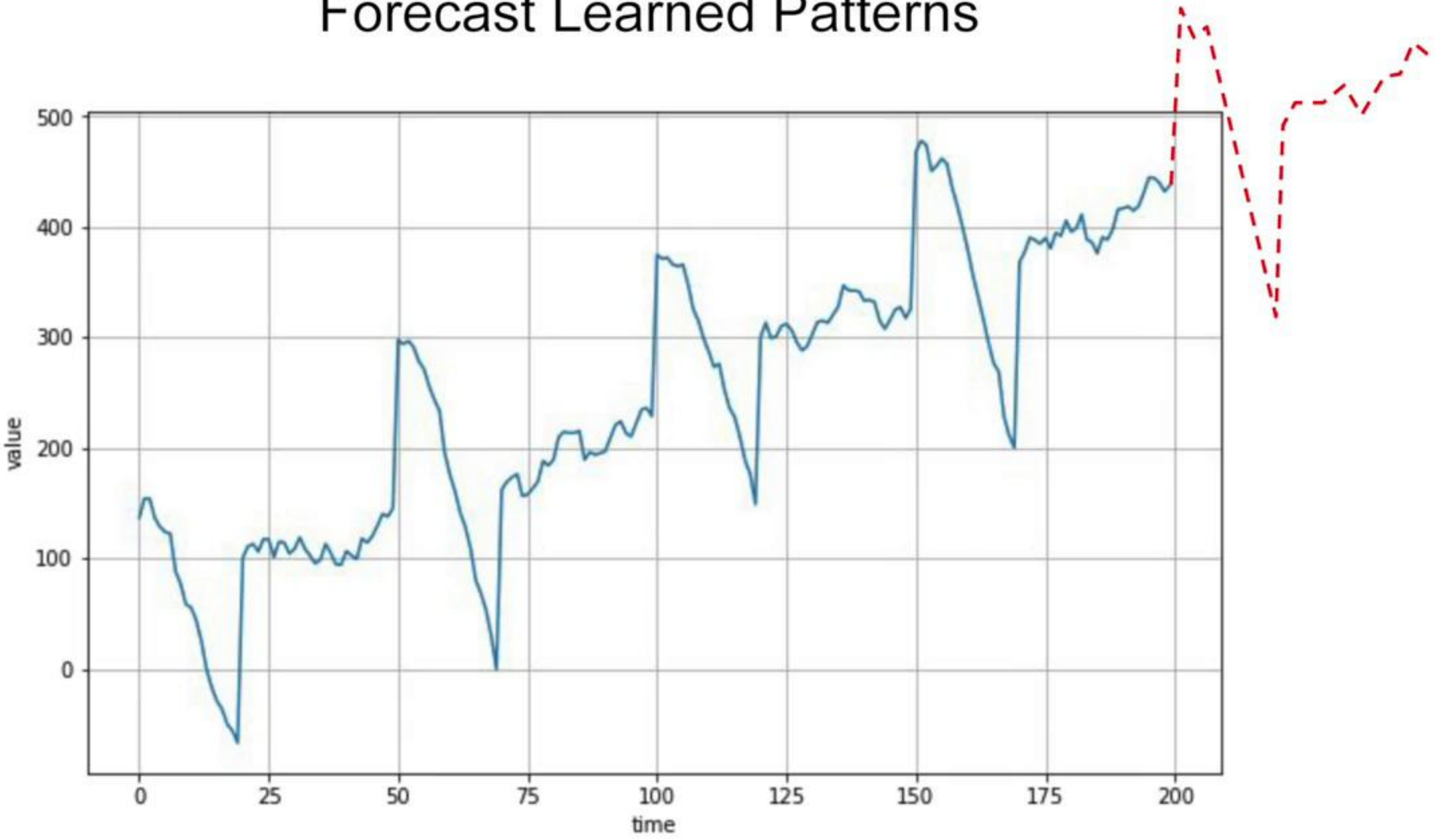
$$v(t) = 0.7 \times v(t-1) + 0.2 \times v(t-50) + \text{occasional spike}$$



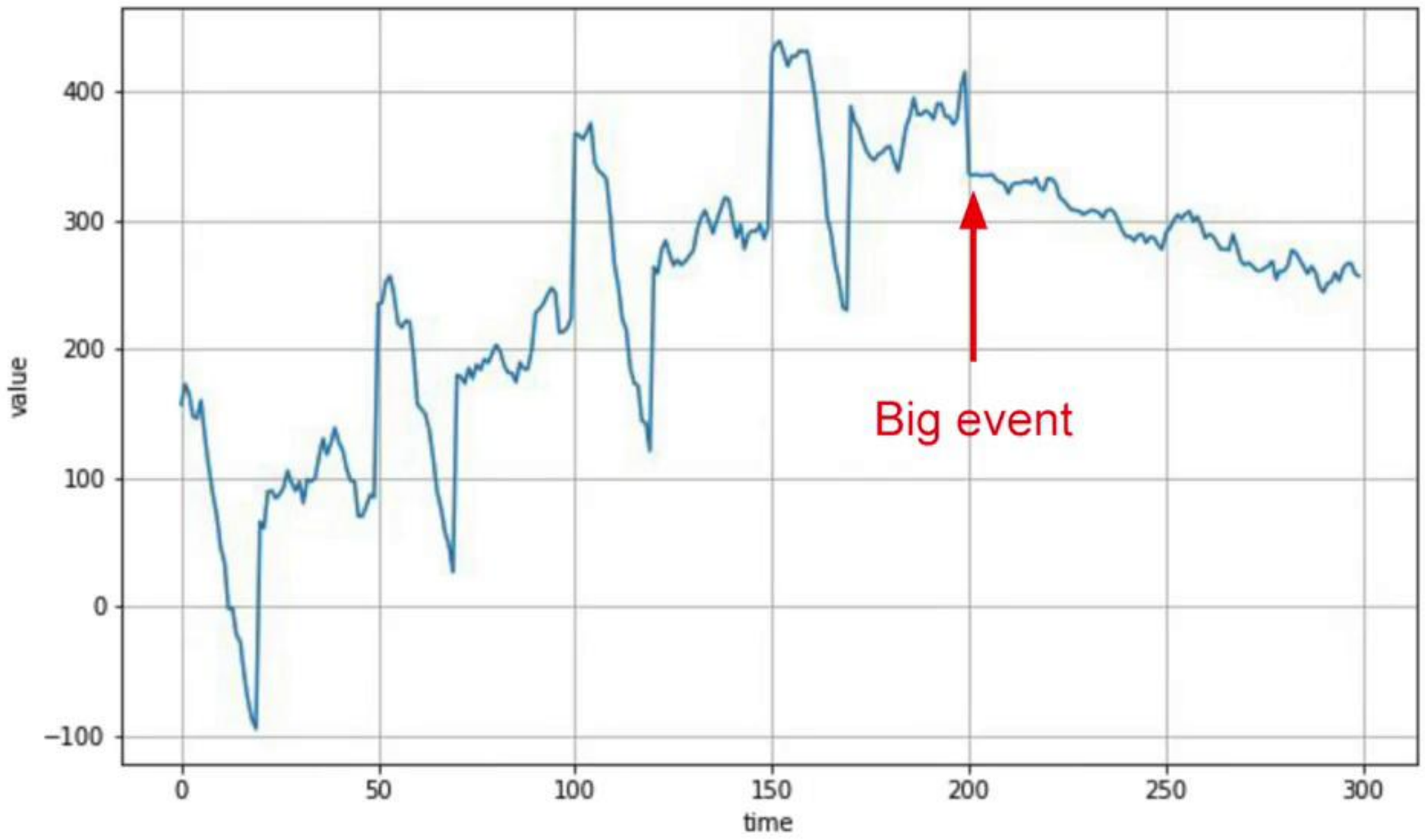
Trend + Seasonality + Autocorrelation + Noise



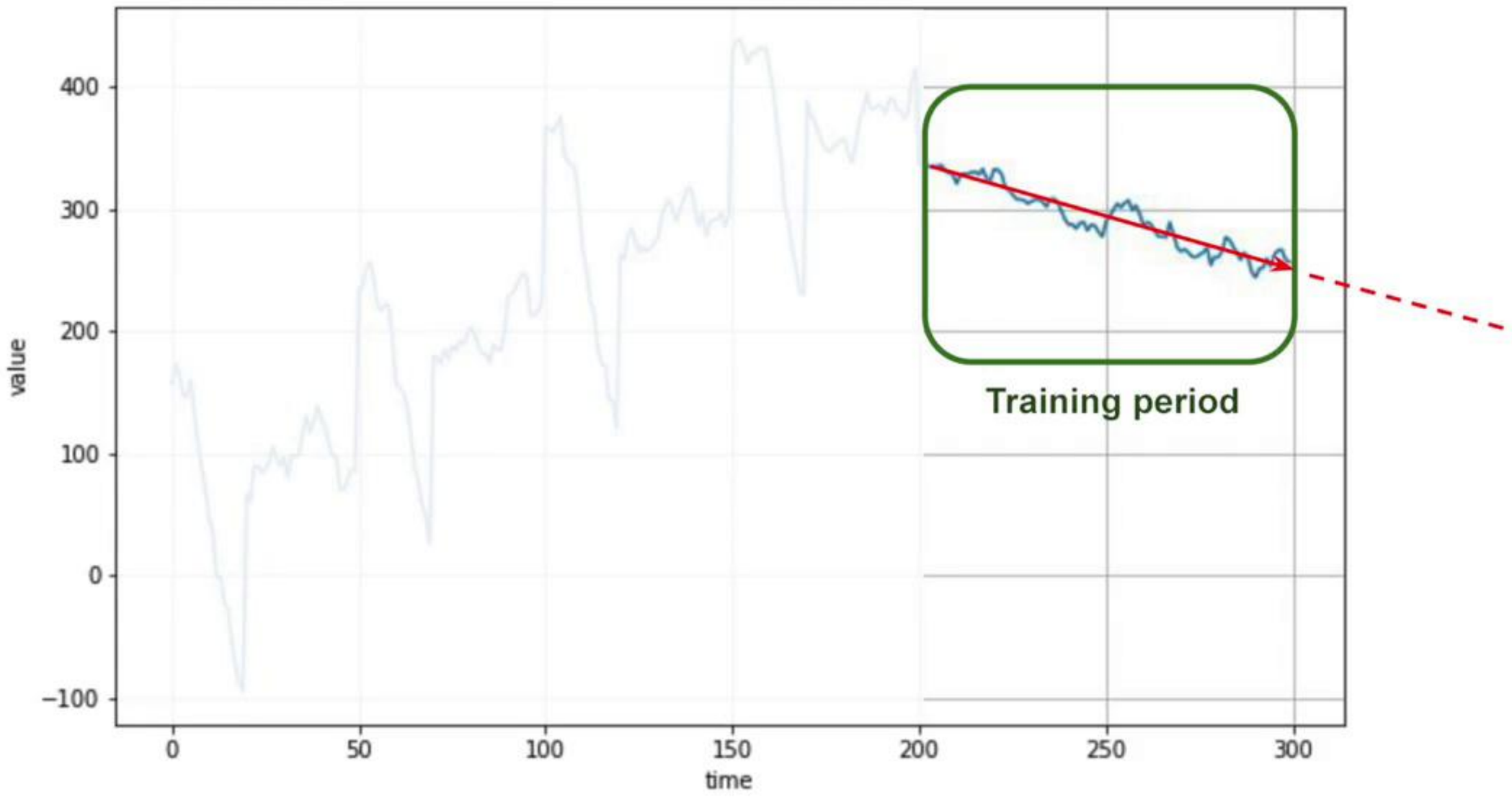
Forecast Learned Patterns



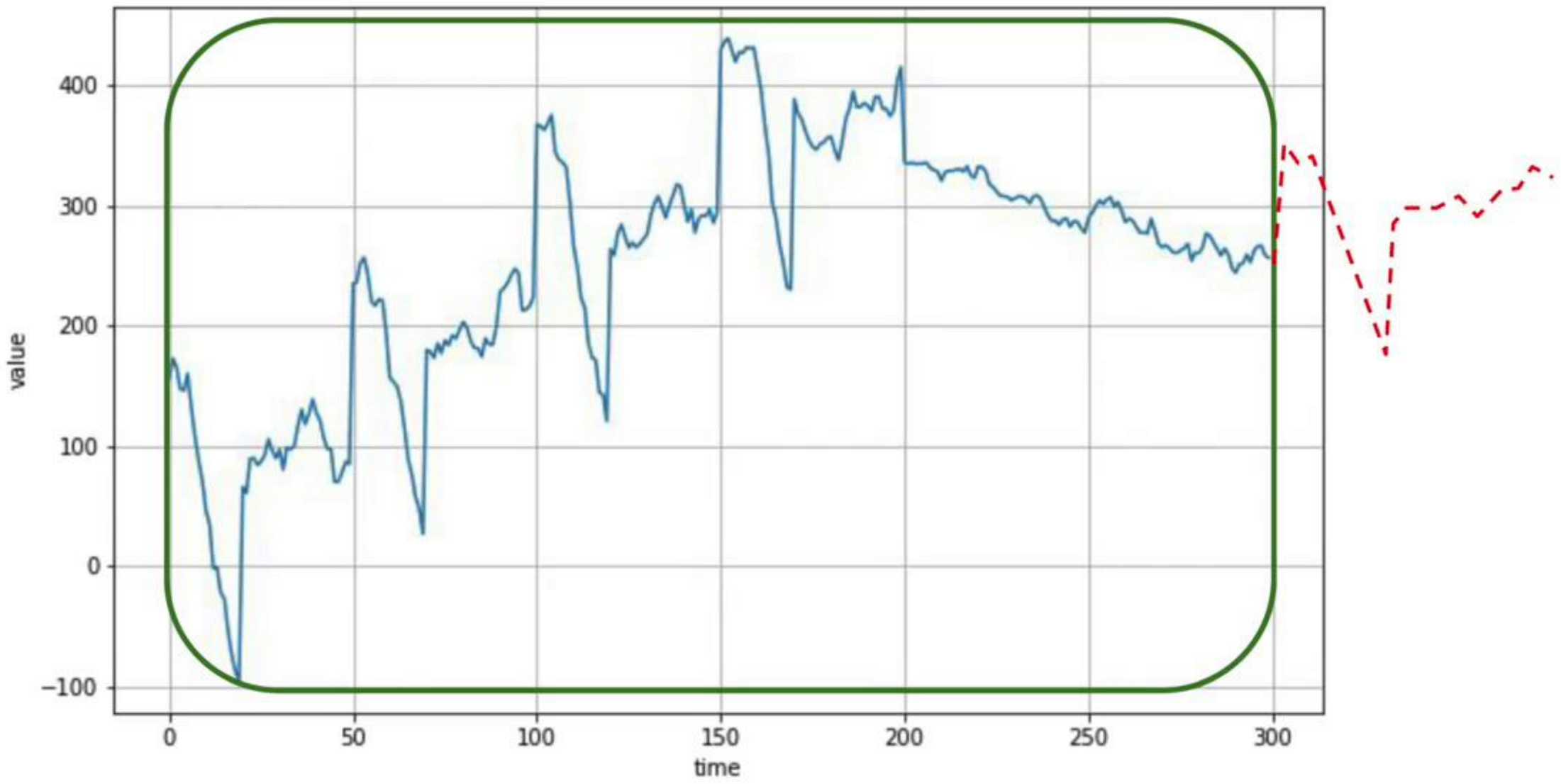
Non-Stationary Time Series



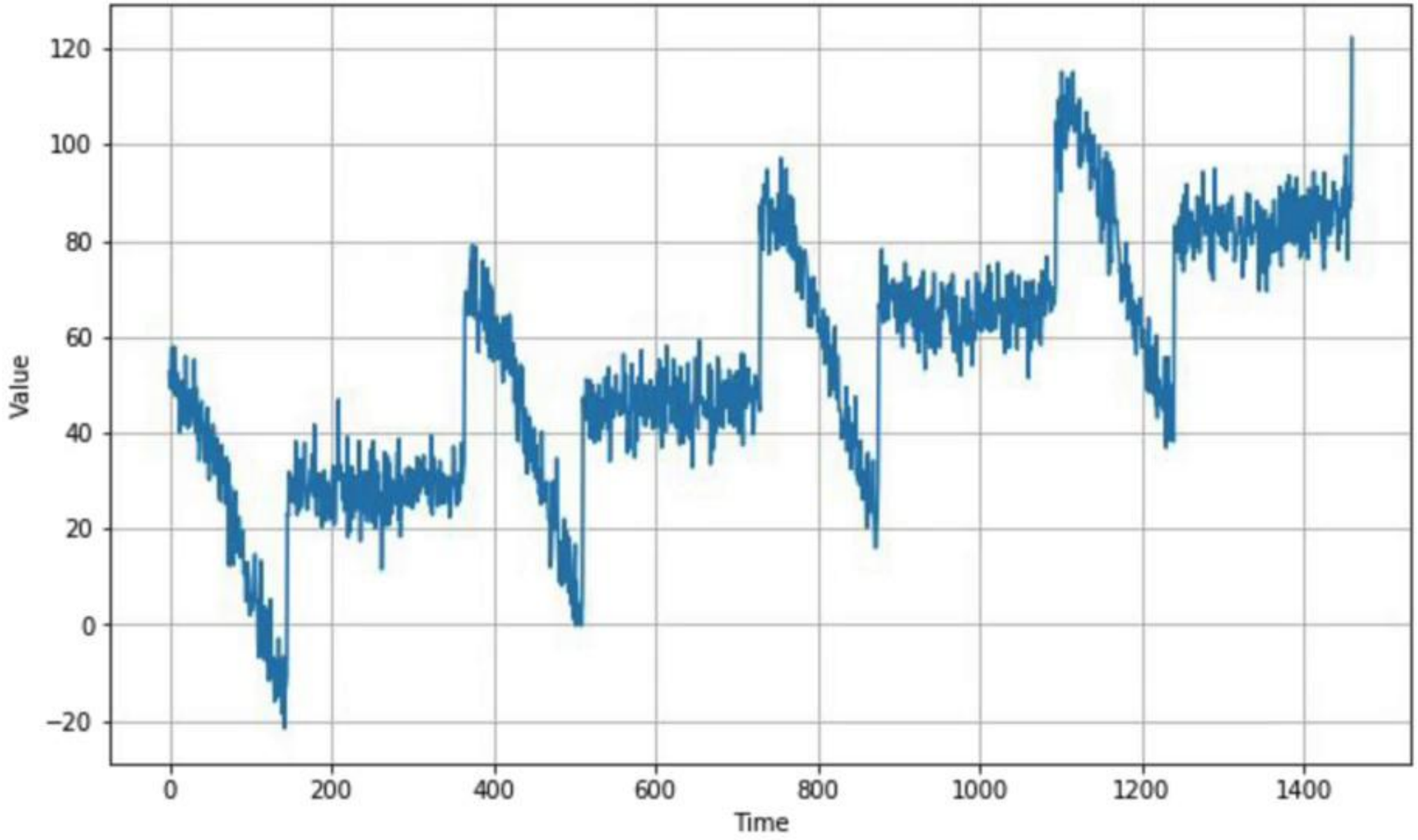
Non-Stationary Time Series



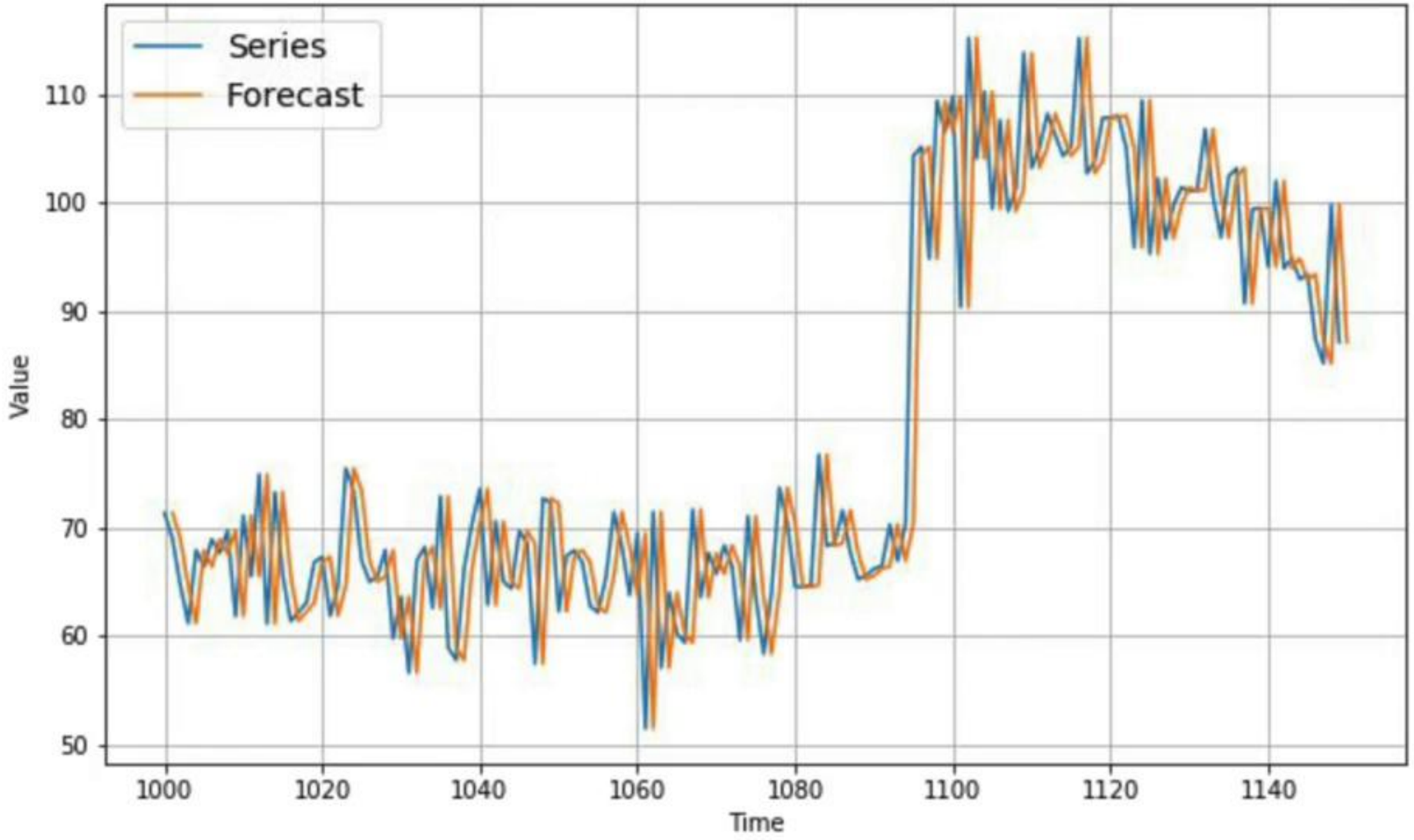
Non-Stationary Time Series



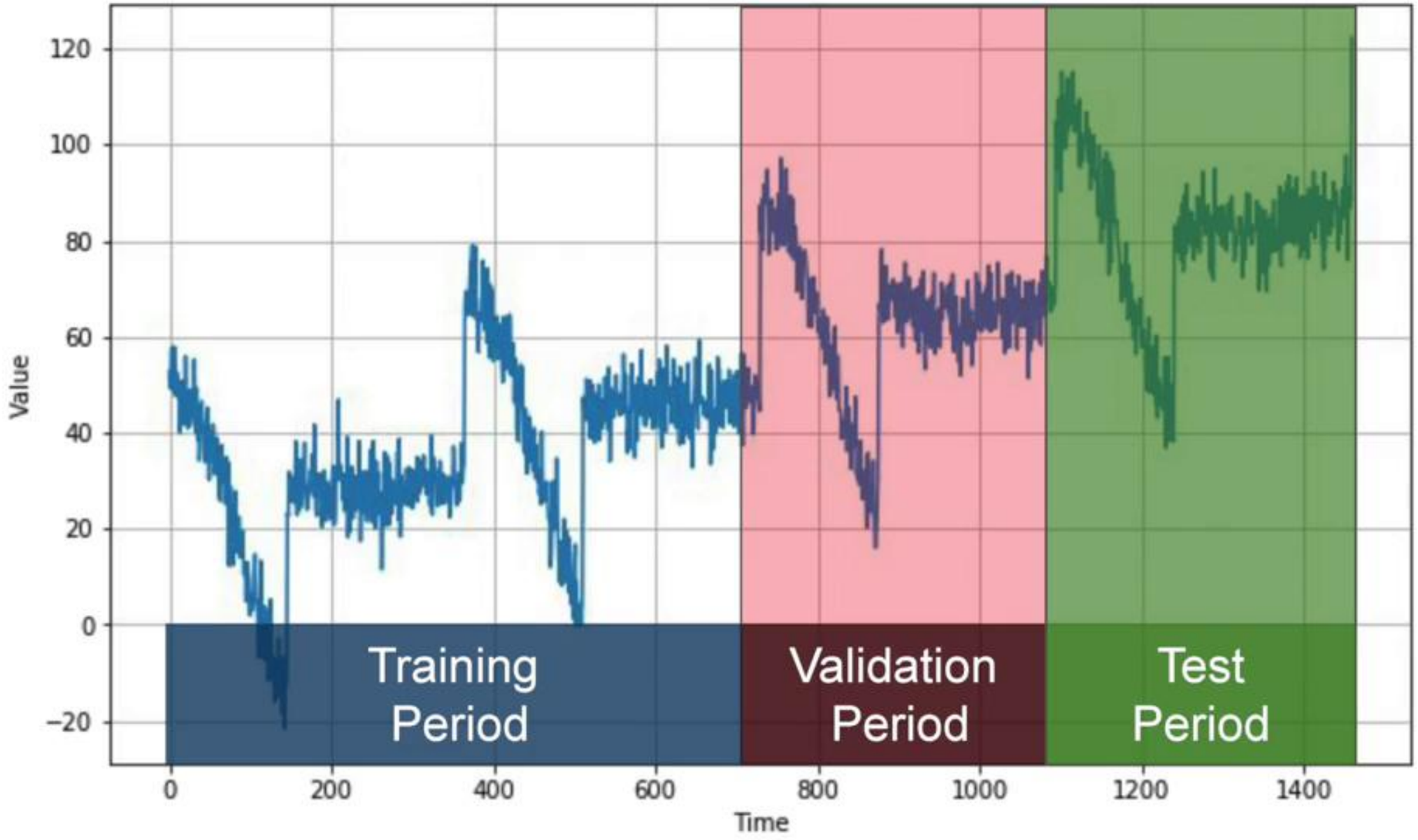
Trend + Seasonality + Noise



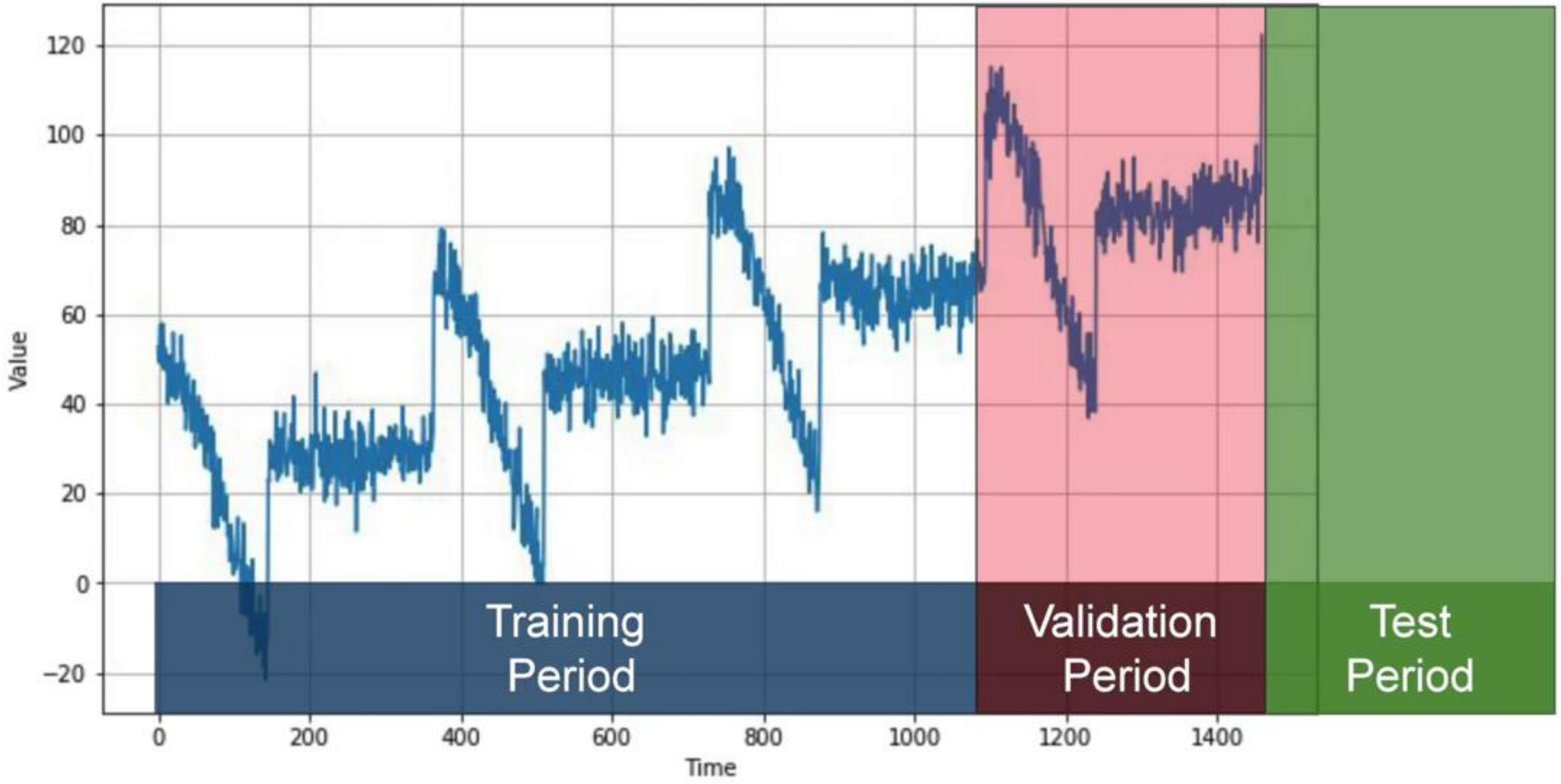
Naive Forecasting



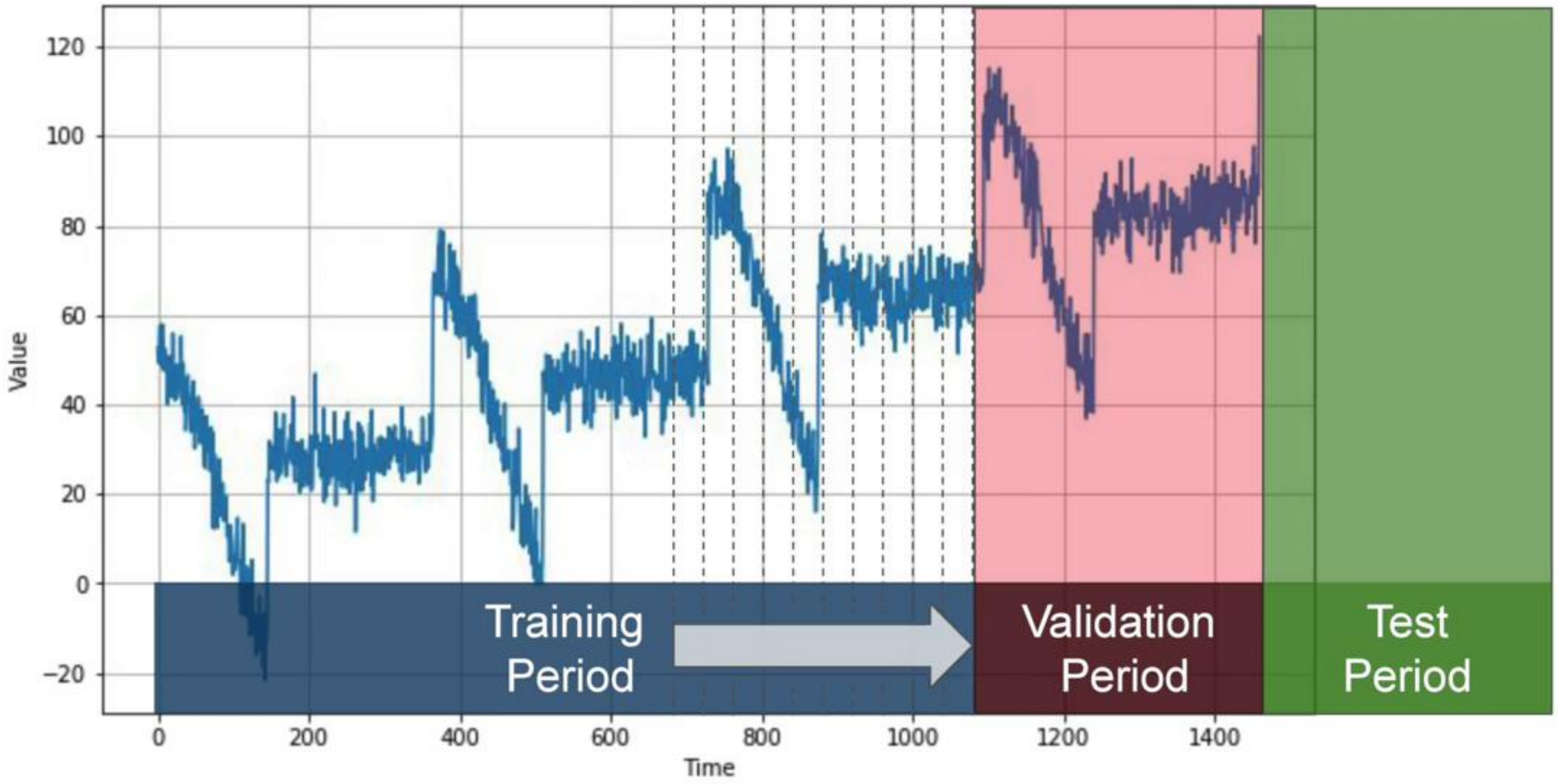
Fixed Partitioning



Fixed Partitioning



Roll-Forward Partitioning



Metrics

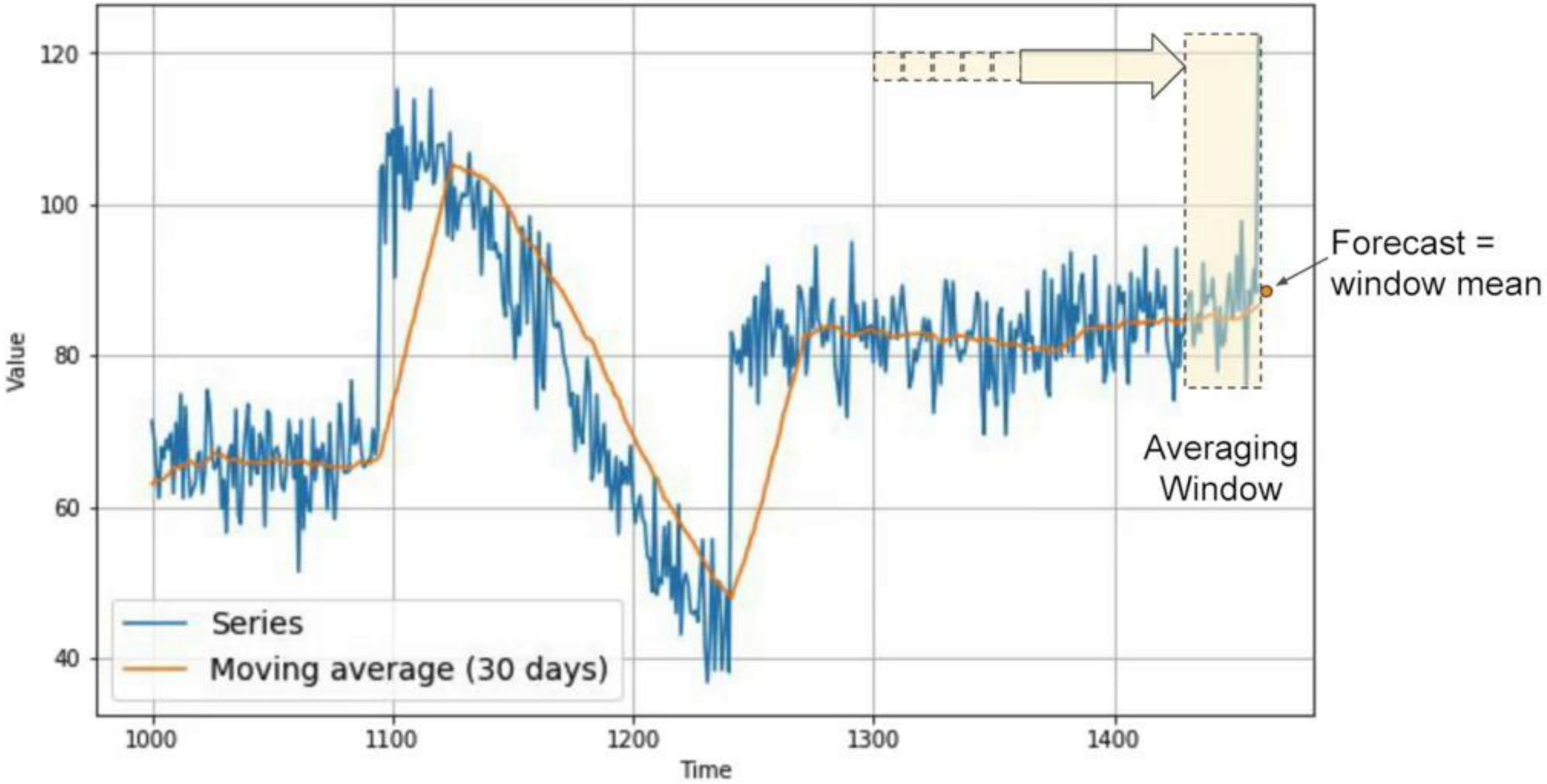
```
errors = forecasts - actual

mse = np.square(errors).mean()

rmse = np.sqrt(mse)

mae = np.abs(errors).mean()

mape = np.abs(errors / x_valid).mean()
```

Naive Forecast MAE

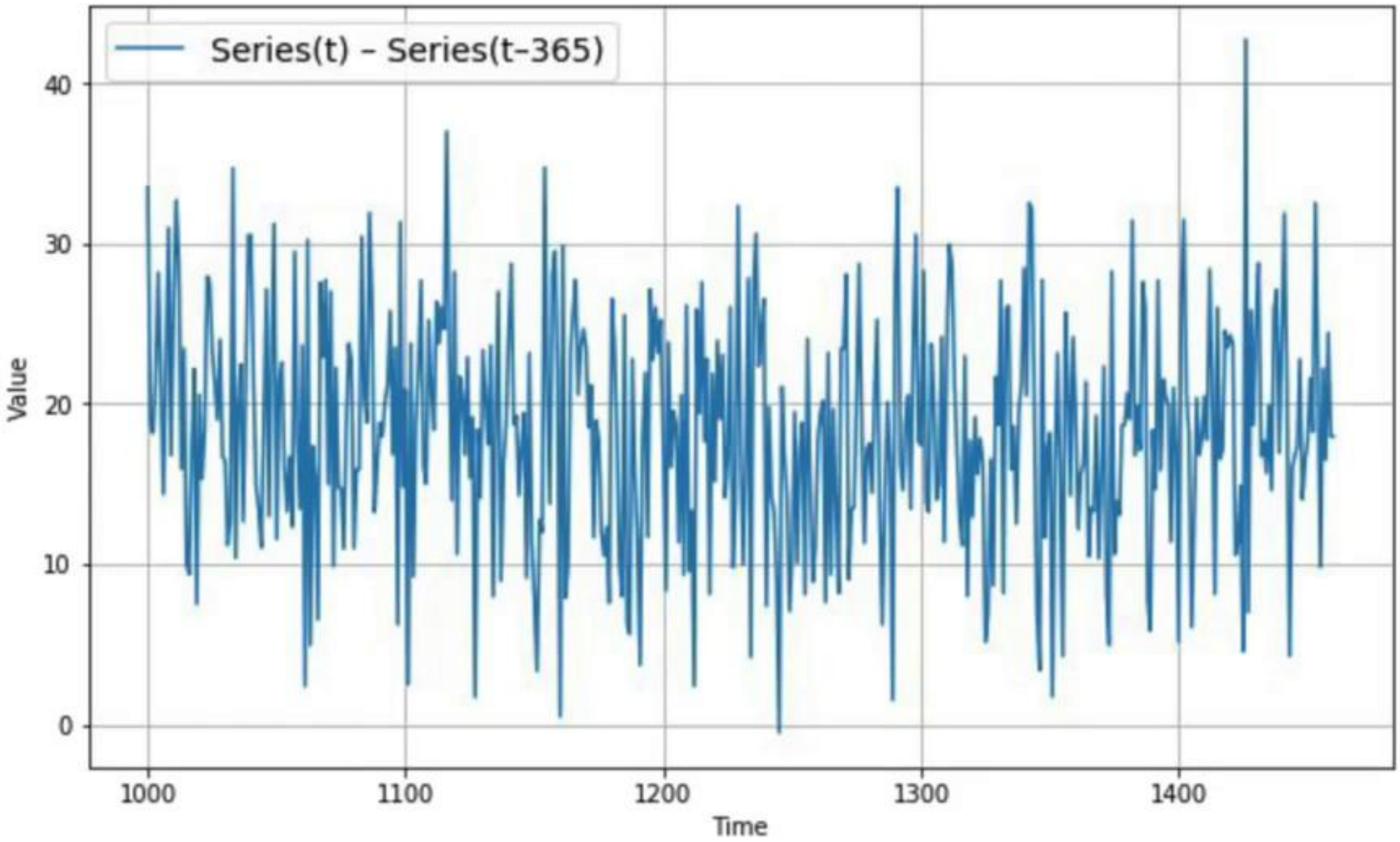
```
keras.metrics.mean_absolute_error(x_valid, naive_forecast).numpy()
```

```
5.937908515321673
```

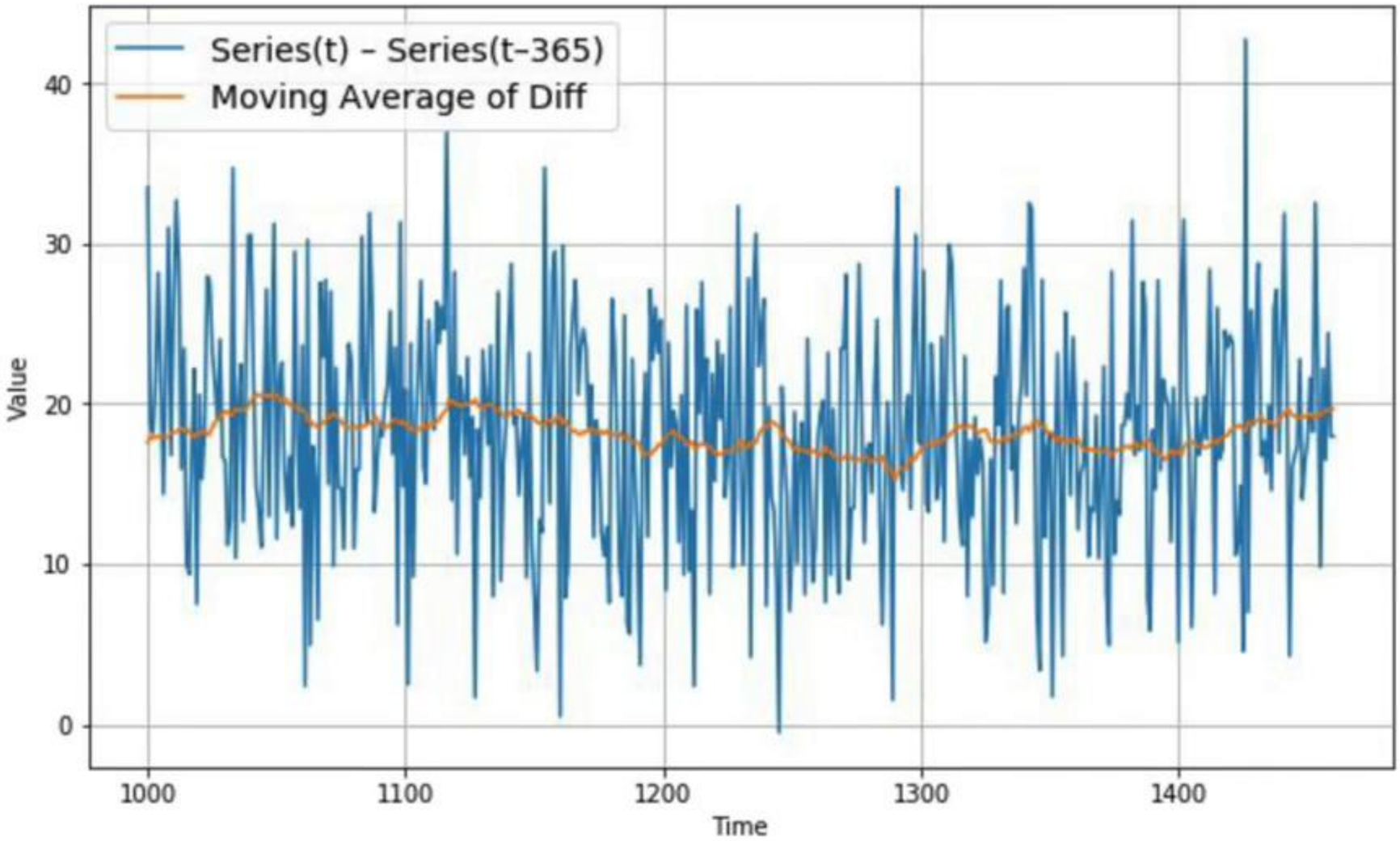
Moving Average



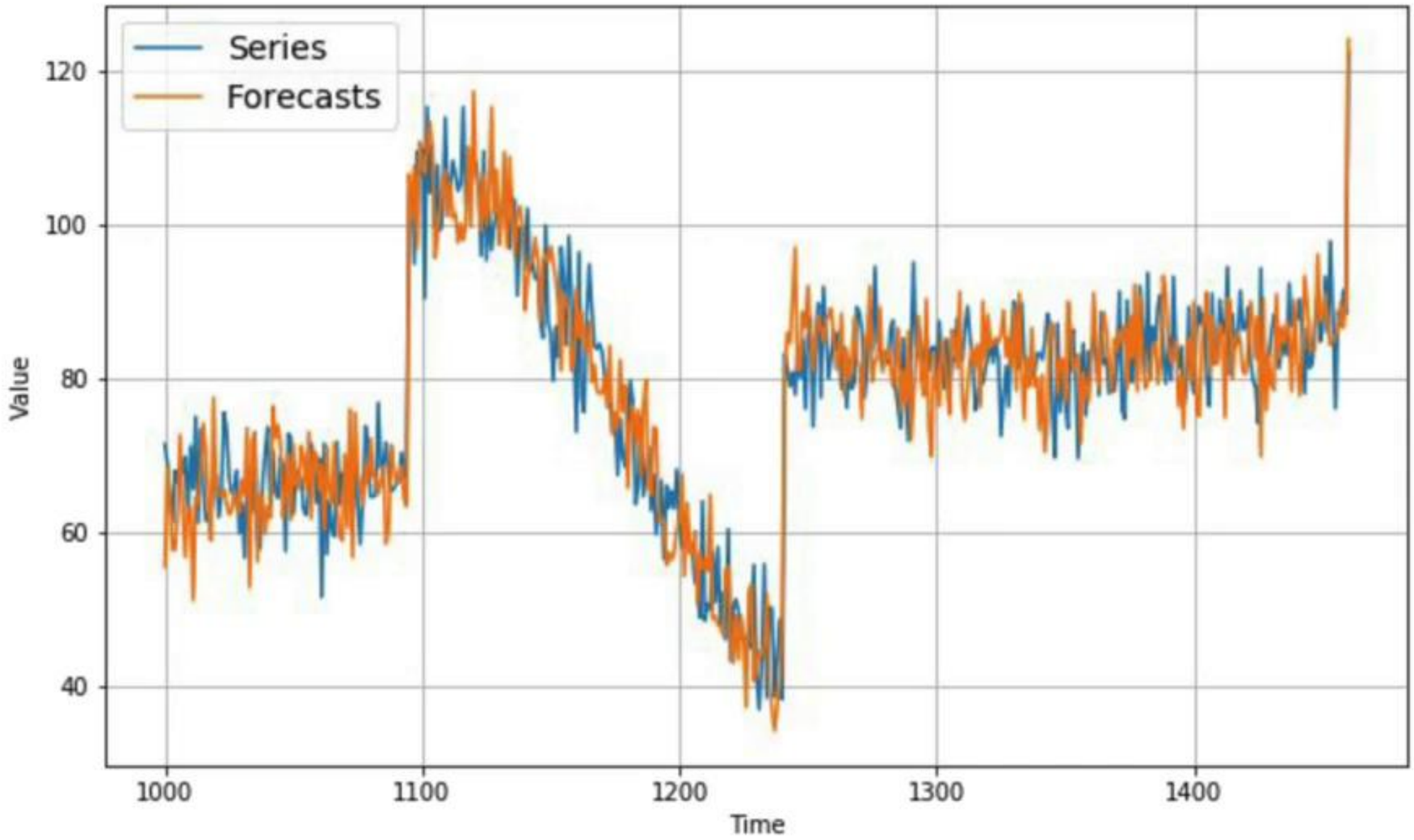
Differencing



Moving Average on Differenced Time Series

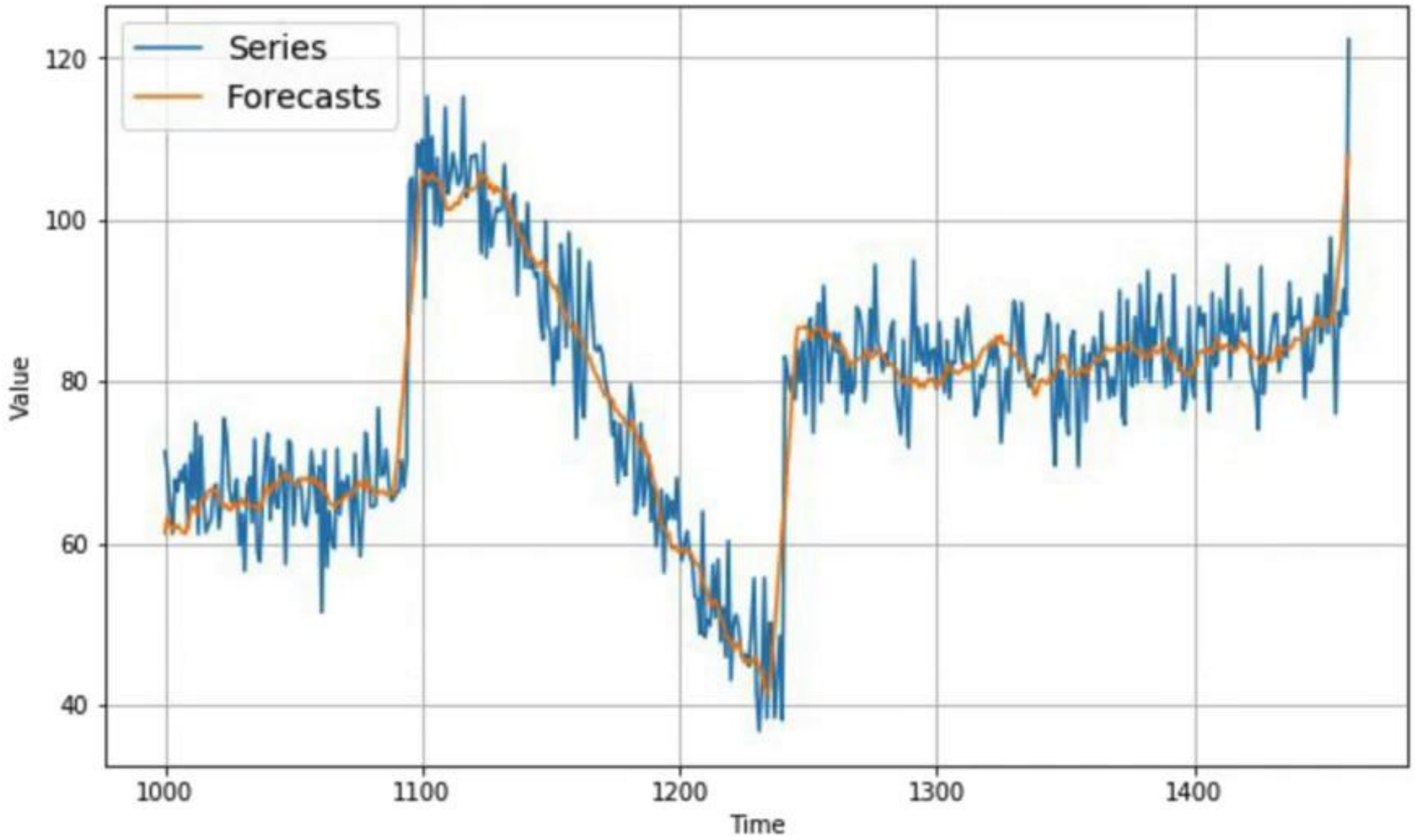


Restoring the Trend and Seasonality



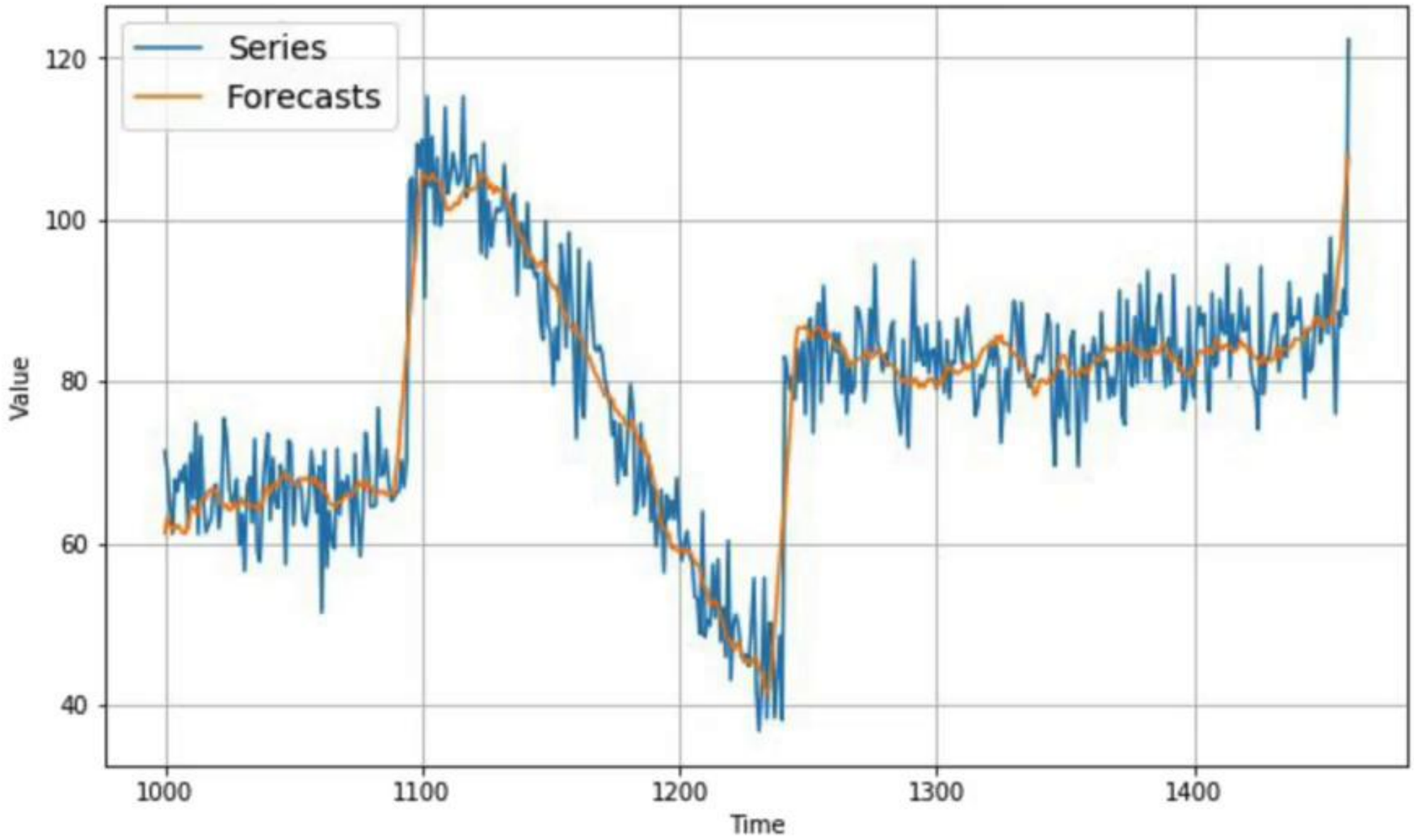
Forecasts = moving average of differenced series + series(t – 365)

Smoothing Both Past and Present Values



Forecasts = trailing moving average of differenced series + centered moving average of past series (t – 365)

Smoothing Both Past and Present Values



Forecasts = trailing moving average of differenced series + centered moving average of past series (t – 365)